FINSIGHT

NovaCrest Bank

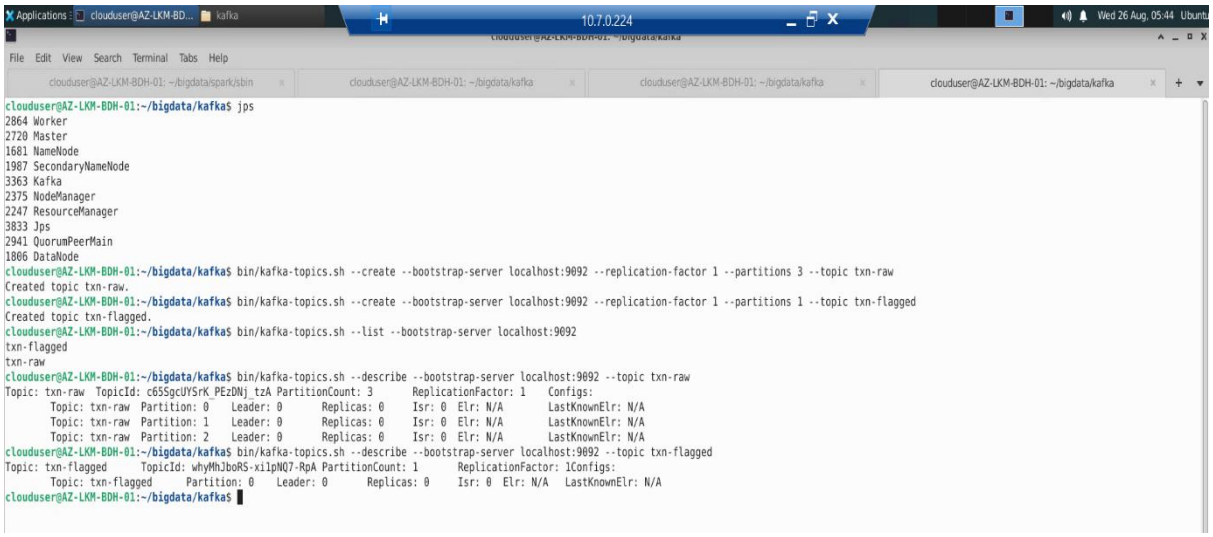
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Day 1 – Data Ingestion

6.1) Kafka Topic Setup

1. Checked Kafka and zookeeper are running or not with the help of JPS.
2. Created Topic txn-raw with 3 Partitions.
3. Created Topic txn-flagged with 1 Partition.
4. Listed the topics
5. Described the topics



--- Finding Transactions.csv

--- Also checking and Verifying

```
clouduser@AZ-LKM-BDH-01:~/bigdata/kafka$ cd /home/clouduser/Downloads/Bigdata-Data-set-file/src-data/
clouduser@AZ-LKM-BDH-01:~/Downloads/Bigdata-Data-set-file/src-data$ ls
Transactions.csv  neo4j_accounts_nodes.csv  neo4j_received_rels.csv  neo4j_sent_rels.csv  neo4j_transaction_nodes.csv  noveacrest_customers.json
clouduser@AZ-LKM-BDH-01:~/Downloads/Bigdata-Data-set-file/src-data$ head -n 7 /home/clouduser/Downloads/Bigdata-Data-set-file/src-data/Transactions.csv
step,type,amount,nameOrig,oldbalanceOrg,newbalanceOrig,nameDest,oldbalanceDest,newbalanceDest,isFraud,isFlaggedFraud
1,PAYMENT,9839.64,C1231006815,170136.0,160296.36,M1979787155,0.0,0.0,0,0
1,PAYMENT,1864.28,C1666544295,21249.0,19384.72,M2044282225,0.0,0.0,0,0
1,TRANSFER,181.0,C1305486145,181.0,0.0,C553264065,0.0,0.0,1,0
1,CASH_OUT,181.0,C840083671,181.0,0.0,C38997010,21182.0,0.0,1,0
1,PAYMENT,11668.14,C2048537720,41554.0,29885.86,M1230701703,0.0,0.0,0,0
1,PAYMENT,7817.71,C90045638,53860.0,46042.29,M573487274,0.0,0.0,0,0
clouduser@AZ-LKM-BDH-01:~/Downloads/Bigdata-Data-set-file/src-data$
```

6.2) Python Kafka Producer

1) Created producer.py file

Path : /home/clouduser/producer.py

1. Read the raw dataset Transactions.csv line by line.
2. Converted CSV text fields into typed variables (int, float, string).
3. Structured each row into a standard JSON message containing:
    - **schema**: Column definitions and data types (int32, double, string).
    - **payload**: The actual transaction values.
4. Sent each message to the txn-raw topic with key = nameOrig.
5. Throttled throughput to **1,000 messages per second** using time.sleep().

2) Ran python3 producer.py

```
Successfully installed kafka-python-3.0.11
clouduser@AZ-LKM-BDH-01:~/capstone$ python3 producer.py
Starting Kafka producer...
Reading: /home/clouduser/Downloads/Bigdata-Dataset/src-data/Transactions.csv
Sending to topic: txn-raw
Target rate: 1000 messages/sec
Sent 10000 messages | Rate: 1000.0 msg/sec
Sent 20000 messages | Rate: 1000.0 msg/sec
Sent 30000 messages | Rate: 1000.0 msg/sec
Sent 40000 messages | Rate: 1000.0 msg/sec
Sent 50000 messages | Rate: 1000.0 msg/sec
Sent 60000 messages | Rate: 1000.0 msg/sec
Sent 70000 messages | Rate: 1000.0 msg/sec
Sent 80000 messages | Rate: 1000.0 msg/sec
Sent 90000 messages | Rate: 1000.0 msg/sec
Sent 100000 messages | Rate: 1000.0 msg/sec
Sent 110000 messages | Rate: 1000.0 msg/sec
Sent 120000 messages | Rate: 1000.0 msg/sec
Sent 130000 messages | Rate: 1000.0 msg/sec
Sent 140000 messages | Rate: 1000.0 msg/sec
Sent 150000 messages | Rate: 1000.0 msg/sec
Sent 160000 messages | Rate: 1000.0 msg/sec
Sent 170000 messages | Rate: 1000.0 msg/sec
Sent 180000 messages | Rate: 1000.0 msg/sec
Sent 190000 messages | Rate: 1000.0 msg/sec
Sent 200000 messages | Rate: 1000.0 msg/sec
```

#Verification

kafka-console-consumer.sh --bootstrap-server localhost:9092 --topic txn-raw --from-beginning --max-messages 5

6.3) HDFS Landing Structure

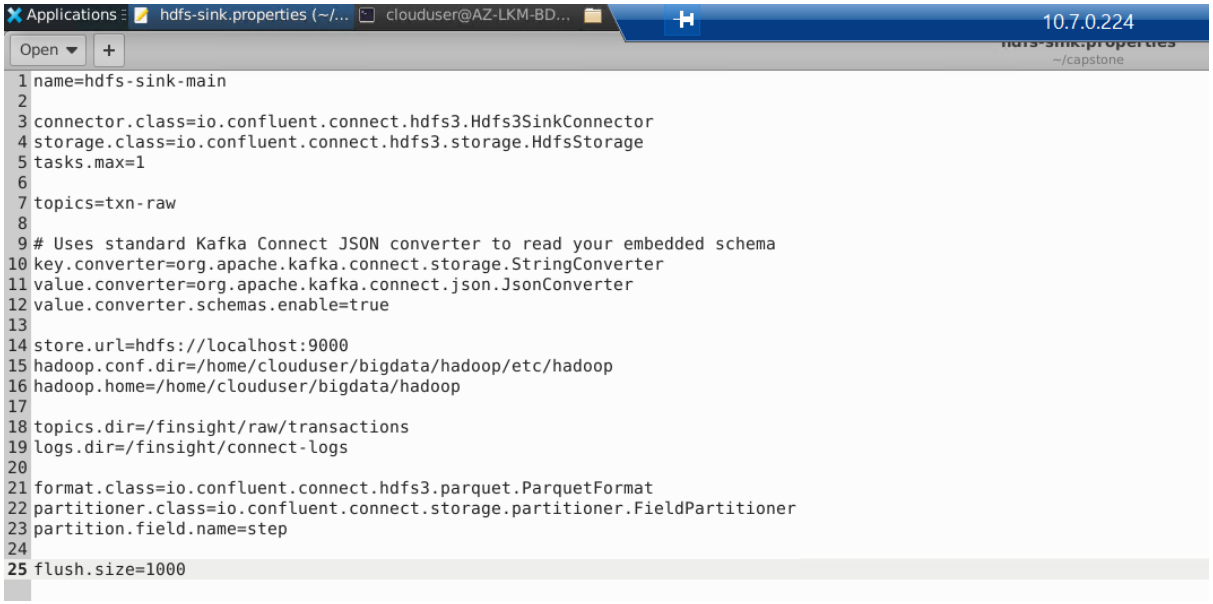
1) Create HDFS Directories

2) verifying Directories

```
clouduser@AZ-LKM-BDH-01:~/bigdata/confluent$ hdfs dfs -mkdir -p /finsight/raw/transactions
clouduser@AZ-LKM-BDH-01:~/bigdata/confluent$ hdfs dfs -mkdir -p /finsight/connect-logs
clouduser@AZ-LKM-BDH-01:~/bigdata/confluent$ hdfs dfs -ls /
Found 5 items
drwxr-xr-x   - clouduser supergroup          0 2026-08-26 09:49 /finsight
drwxr-xr-x   - clouduser supergroup          0 2025-11-30 09:22 /hbase
drwxr-xr-x   - clouduser supergroup          0 2026-08-24 08:44 /home
drwx------   - clouduser supergroup          0 2026-08-20 05:15 /tmp
drwxr-xr-x   - clouduser supergroup          0 2026-07-21 06:25 /user
clouduser@AZ-LKM-BDH-01:~/bigdata/confluent$
```

2) Created the HDFS Sink Configuration

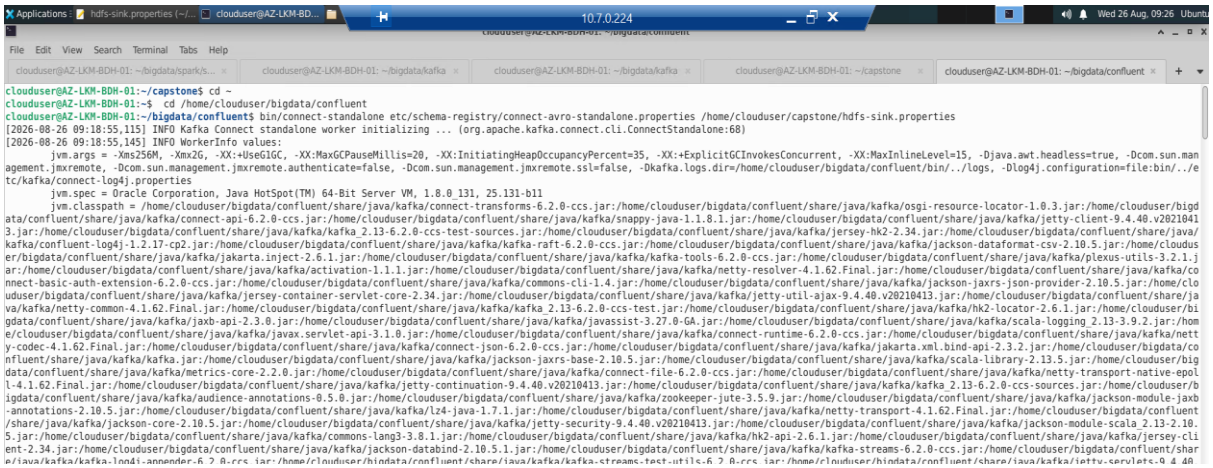
Path : /home/clouduser/capstone/hdfs-sink.properties



3)Start Kafka connect

1. **cd /home/clouduser/bigdata/confluent**

2. bin/connect-standalone etc/schema-registry/connect-avro-standalone.properties /home/clouduser/capstone/hdfs-sink.properties

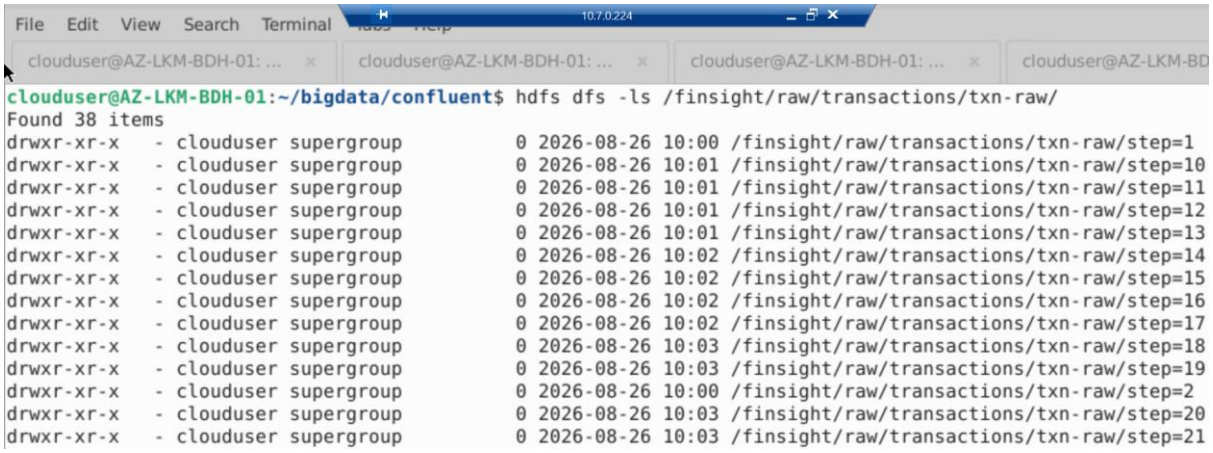


1. View all step partitions:

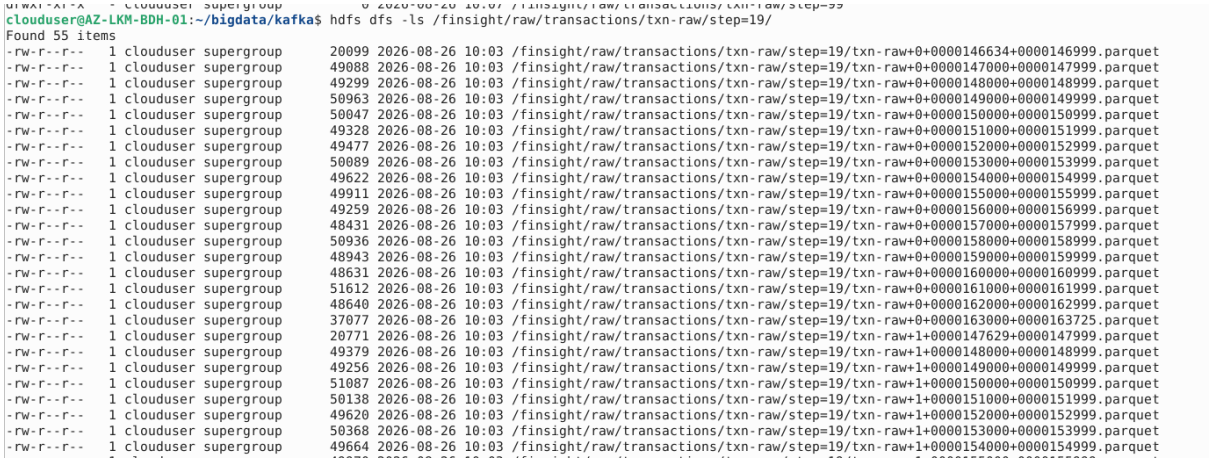
hdfs dfs -ls /finsight/raw/transactions/txn-raw/

2. View the Parquet file inside a partition (e.g. step=1 or step=19):

hdfs dfs -ls /finsight/raw/transactions/txn-raw/step=19/



#3 Parquet data files stored inside that folder



Day 2 – Spark Processing

7.1) Spark Streaming — Real-Time Fraud Detection

1)Create HDFS Folders for Checkpoints & Metrics

hdfs dfs -mkdir -p /finsight/checkpoints/fraud

hdfs dfs -mkdir -p /finsight/processed/streaming_metrics



2)Create fraud_detection.py

3) run

spark-submit --packages org.apache.spark:spark-sql-kafka-0-10_2.12:3.3.0 /home/clouduser/capstone/fraud_detection.py

```
Batch 0 | Total: 49,999 | Flagged: 289 | Fraud Rate: 0.5780%
Batch 1 | Total: 49,999 | Flagged: 326 | Fraud Rate: 0.6520%
Batch 2 | Total: 49,999 | Flagged: 160 | Fraud Rate: 0.3200%
Batch 3 | Total: 49,999 | Flagged: 214 | Fraud Rate: 0.4280%
Batch 4 | Total: 49,999 | Flagged: 182 | Fraud Rate: 0.3640%
Batch 5 | Total: 49,999 | Flagged: 135 | Fraud Rate: 0.2700%
Batch 6 | Total: 49,999 | Flagged: 134 | Fraud Rate: 0.2680%
Batch 7 | Total: 49,999 | Flagged: 67 | Fraud Rate: 0.1340%
Batch 8 | Total: 49,999 | Flagged: 104 | Fraud Rate: 0.2080%
Batch 9 | Total: 49,999 | Flagged: 80 | Fraud Rate: 0.1600%
Batch 10 | Total: 49,999 | Flagged: 14 | Fraud Rate: 0.0280%
Batch 11 | Total: 49,999 | Flagged: 60 | Fraud Rate: 0.1200%
Batch 12 | Total: 49,999 | Flagged: 85 | Fraud Rate: 0.1700%
Batch 13 | Total: 49,999 | Flagged: 52 | Fraud Rate: 0.1040%
Batch 14 | Total: 49,999 | Flagged: 35 | Fraud Rate: 0.0700%
Batch 15 | Total: 49,999 | Flagged: 67 | Fraud Rate: 0.1340%
Batch 16 | Total: 49,999 | Flagged: 30 | Fraud Rate: 0.0600%
Batch 17 | Total: 49,999 | Flagged: 39 | Fraud Rate: 0.0780%
Batch 18 | Total: 49,999 | Flagged: 20 | Fraud Rate: 0.0400%
Batch 19 | Total: 49,999 | Flagged: 15 | Fraud Rate: 0.0300%
Batch 20 | Total: 49,999 | Flagged: 207 | Fraud Rate: 0.4140%
```

4) Verification:

Check Flagged Fraud Records in txn-flagged Kafka Topic

bin/kafka-console-consumer.sh --bootstrap-server localhost:9092 --topic txn-flagged --from-beginning --max-messages 2

```
clouduser@AZ-LKM-BDH-01:~$ cd /home/clouduser/bigdata/kafka/
clouduser@AZ-LKM-BDH-01:~/bigdata/kafka$ bin/kafka-console-consumer.sh --bootstrap-server localhost:9092 --topic txn-flagged --from-beginning --max-messages 2
{"step": 1, "type": "CASH_OUT", "amount": 344464.4, "nameOrig": "C793293778", "oldbalanceOrg": 0.0, "newbalanceOrig": 0.0, "nameDest": "C766572210", "oldbalanceDest": 1133312.56, "newbalanceDest": 0.0, "isFraud"
: 0, "isFlaggedFraud": 0, "fraud_detected": 1, "detected_at": "2026-09-01 04:26:25.280000"}
{"step": 1, "type": "TRANSFER", "amount": 420532.33, "nameOrig": "C582300198", "oldbalanceOrg": 5386.08, "newbalanceOrig": 0.0, "nameDest": "C1971489295", "oldbalanceDest": 614566.02, "newbalanceDest": 0.0, "isF
raud": 0, "isFlaggedFraud": 0, "fraud_detected": 1, "detected_at": "2026-09-01 04:26:25.280000"}
Processed a total of 2 messages
```

5) Check the Streaming Metrics in HDFS:

hdfs dfs -ls /finsight/processed/streaming_metrics

```
clouduser@AZ-LKM-BDH-01:~$ hdfs dfs -ls /finsight/processed/streaming_metrics
Found 257 items
-rw-r--r--   1 clouduser supergroup          0 2026-09-01 04:41 /finsight/processed/streaming_metrics/_SUCCESS
-rw-r--r--   1 clouduser supergroup        725 2026-09-01 04:29 /finsight/processed/streaming_metrics/part-00000-00d24ace-3c63-4e71-b51c-582521c8d651-c000.snappy.parquet
-rw-r--r--   1 clouduser supergroup        725 2026-09-01 04:36 /finsight/processed/streaming_metrics/part-00000-03364828-5e54-4550-ad74-fe115fc66595-c000.snappy.parquet
-rw-r--r--   1 clouduser supergroup        725 2026-09-01 04:35 /finsight/processed/streaming_metrics/part-00000-038a7dda-b038-4827-bf09-dba98d3a93ad-c000.snappy.parquet
-rw-r--r--   1 clouduser supergroup        725 2026-09-01 04:34 /finsight/processed/streaming_metrics/part-00000-039e3106-bad7-4ea7-a49f-a398d4787cd4-c000.snappy.parquet
-rw-r--r--   1 clouduser supergroup        725 2026-09-01 04:27 /finsight/processed/streaming_metrics/part-00000-053d3e46-2454-49bc-a028-cd9110dec652-c000.snappy.parquet
-rw-r--r--   1 clouduser supergroup        725 2026-09-01 04:41 /finsight/processed/streaming_metrics/part-00000-060512bb-7804-4065-97cc-e97aa4ef71f5-c000.snappy.parquet
-rw-r--r--   1 clouduser supergroup        725 2026-09-01 04:32 /finsight/processed/streaming_metrics/part-00000-09ecec33-de39-41db-8358-e66d964eeec0-c000.snappy.parquet
-rw-r--r--   1 clouduser supergroup        725 2026-09-01 04:38 /finsight/processed/streaming_metrics/part-00000-0e76ab53-3164-42e1-99ba-2b9e9f58ea8c-c000.snappy.parquet
-rw-r--r--   1 clouduser supergroup        725 2026-09-01 04:31 /finsight/processed/streaming_metrics/part-00000-1043dd31-5084-4833-b2a4-e4ec7b17db05-c000.snappy.parquet
-rw-r--r--   1 clouduser supergroup        725 2026-09-01 04:37 /finsight/processed/streaming_metrics/part-00000-1080e146-724a-44b5-8137-95f43f3a14d2-c000.snappy.parquet
```

6) count the total flagged fraud records

bin/kafka-console-consumer.sh --bootstrap-server localhost:9092 --topic txn-flagged --from-beginning --timeout-ms 5000 | wc -l

```
clouduser@AZ-LKM-BDH-01:~/bigdata/kafka$ bin/kafka-console-consumer.sh --bootstrap-server localhost:9092 --topic txn-flagged --from-beginning --timeout-ms 5000 | wc -l
[2026-09-01 07:38:25,111] ERROR Error processing message, terminating consumer process:  (org.apache.kafka.tools.consumer.ConsoleConsumer)
org.apache.kafka.common.errors.TimeoutException
Processed a total of 5600 messages
5600
```

7.2) Spark Streaming — Real-Time Customer Churn Detection

STEP 1: 1) Create the txn-churn Kafka Topic

bin/kafka-topics.sh --bootstrap-server localhost:9092 --create --topic txn-churn --partitions 1 --replication-factor 1

2) Verify it exists

bin/kafka-topics.sh --bootstrap-server localhost:9092 --describe --topic txn-churn

```
clouduser@AZ-LKM-BDH-01:~/bigdata/kafka$ bin/kafka-topics.sh --bootstrap-server localhost:9092 --create --topic txn-churn --partitions 1 --replication-factor 1
Created topic txn-churn.
clouduser@AZ-LKM-BDH-01:~/bigdata/kafka$ bin/kafka-topics.sh --bootstrap-server localhost:9092 --describe --topic txn-churn
Topic: txn-churn        TopicId: _SzKs4miRQq91C5mABaUmg PartitionCount: 1       ReplicationFactor: 1    Configs:
        Topic: txn-churn        Partition: 0    Leader: 0       Replicas: 0     Isr: 0  Elr: N/A        LastKnownElr: N/A
clouduser@AZ-LKM-BDH-01:~/bigdata/kafka$
```

STEP 2: Create HDFS Folders for Churn Alerts & Checkpoints

hdfs dfs -mkdir -p /finsight/processed/churn_alerts

hdfs dfs -mkdir -p /finsight/checkpoints/churn

```
clouduser@AZ-LKM-BDH-01:~$ hdfs dfs -mkdir -p /finsight/processed/churn_alerts
clouduser@AZ-LKM-BDH-01:~$ hdfs dfs -mkdir -p /finsight/checkpoints/churn
clouduser@AZ-LKM-BDH-01:~$ hdfs dfs -ls /finsight/processed
Found 2 items
drwxr-xr-x   - clouduser supergroup          0 2026-09-01 07:26 /finsight/processed/churn_alerts
drwxr-xr-x   - clouduser supergroup          0 2026-09-01 04:41 /finsight/processed/streaming_metrics
clouduser@AZ-LKM-BDH-01:~$ hdfs dfs -ls /finsight/checkpoints
Found 2 items
drwxr-xr-x   - clouduser supergroup          0 2026-09-01 07:26 /finsight/checkpoints/churn
drwxr-xr-x   - clouduser supergroup          0 2026-08-31 09:38 /finsight/checkpoints/fraud
clouduser@AZ-LKM-BDH-01:~$
```

STEP 3: Create the churn_detection.py

nano /home/clouduser/capstone/churn_detection.py

STEP 4: Run the Churn Detection Job

spark-submit --packages org.apache.spark:spark-sql-kafka-0-10_2.12:3.3.0 /home/clouduser/capstone/churn_detection.py

```
=====================================================
       CHURN STREAMING MICRO-BATCH : 0
=====================================================
  Total Customers Evaluated : 49,999
  Customers Flagged at Risk : 15,488
  Output Kafka Topic        : txn-churn
  HDFS Alert Destination    : /finsight/processed/churn_alerts
=====================================================


=====================================================
       CHURN STREAMING MICRO-BATCH : 1
=====================================================
  Total Customers Evaluated : 49,999
  Customers Flagged at Risk : 17,957
  Output Kafka Topic        : txn-churn
  HDFS Alert Destination    : /finsight/processed/churn_alerts
=====================================================


=====================================================
       CHURN STREAMING MICRO-BATCH : 2
=====================================================
  Total Customers Evaluated : 49,999
  Customers Flagged at Risk : 18,510
  Output Kafka Topic        : txn-churn
  HDFS Alert Destination    : /finsight/processed/churn_alerts
=====================================================
```

Verify :

bin/kafka-console-consumer.sh --bootstrap-server localhost:9092 --topic txn-churn --from-beginning --max-messages 3

```
clouduser@AZ-LKM-BDH-01:~/bigdata/kafka$ bin/kafka-console-consumer.sh --bootstrap-server localhost:9092 --topic txn-churn --from-beginning --max-messages 3
{"customerId": "C855700733", "txn_count": 1, "window_start_step": 1, "window_end_step": 1, "avg_amount": 82685.93, "cashout_count": 1, "spend_count": 0, "low_balance_count": 1, "s1": 1, "s2": 0, "s3": 1, "s4": 0
, "total_signals": 2, "triggered_signals": "", "alert_timestamp": "2026-09-01 08:19:30.553000"}
{"customerId": "C1898629094", "txn_count": 1, "window_start_step": 2, "window_end_step": 2, "avg_amount": 158934.62, "cashout_count": 1, "spend_count": 0, "low_balance_count": 1, "s1": 1, "s2": 0, "s3": 1, "s4":
 0, "total_signals": 2, "triggered_signals": "", "alert_timestamp": "2026-09-01 08:19:30.553000"}
{"customerId": "C1240536793", "txn_count": 1, "window_start_step": 8, "window_end_step": 8, "avg_amount": 18359.09, "cashout_count": 1, "spend_count": 0, "low_balance_count": 1, "s1": 1, "s2": 0, "s3": 1, "s4":
0, "total_signals": 2, "triggered_signals": "", "alert_timestamp": "2026-09-01 08:19:30.553000"}
Processed a total of 3 messages
```

7.3) Spark Core — Batch Risk Scoring

STEP 1: Create HDFS Folders for Outputs

hdfs dfs -mkdir -p /finsight/processed/risk_scores

hdfs dfs -mkdir -p /finsight/processed/daily_summary

```
clouduser@AZ-LKM-BDH-01:~$      hdfs dfs -mkdir -p /finsight/processed/risk_scores
clouduser@AZ-LKM-BDH-01:~$ hdfs dfs -mkdir -p /finsight/processed/daily_summary
clouduser@AZ-LKM-BDH-01:~$ hdfs dfs -ls /finsight/processed
Found 4 items
drwxr-xr-x   - clouduser supergroup          0 2026-09-01 09:08 /finsight/processed/churn_alerts
drwxr-xr-x   - clouduser supergroup          0 2026-09-01 09:09 /finsight/processed/daily_summary
drwxr-xr-x   - clouduser supergroup          0 2026-09-01 09:08 /finsight/processed/risk_scores
drwxr-xr-x   - clouduser supergroup          0 2026-09-01 04:41 /finsight/processed/streaming_metrics
clouduser@AZ-LKM-BDH-01:~$
```

STEP 2: Create the risk_scoring.py Script

nano /home/clouduser/capstone/batch_risk_scoring.py

STEP 3: Run the Batch Risk Scoring Job

spark-submit /home/clouduser/capstone/risk_scoring.py

```
============================================================
     TASK 7.3 BATCH RISK SCORING COMPLETED SUCCESSFULLY
============================================================
  Total Customers Scored  : 199,779
  Low Risk (< 0.25)       : 135,573
  Medium Risk (0.25-0.6)  : 64,203
  High Risk (> 0.60)      : 3
------------------------------------------------------------
  Daily Summary Rows (R2) : 2,674
============================================================
```

```
>>> OUTPUT 1 SAMPLE: CUSTOMER RISK SCORES & 4 FACTORS <<<
+-----------+--------------------+----------------------+------------------+------------------+----------+---------+
|customerId |transaction_frequency|average_transfer_amount|cash_out_proportion|unique_destinations|risk_score|risk_tier|
+-----------+--------------------+----------------------+------------------+------------------+----------+---------+
|C652197148 |1                   |160839.21             |1.0000            |1                 |0.2526    |Medium   |
|C1589340867|1                   |4535.38               |0.0000            |1                 |0.0001    |Low      |
|C1463997315|1                   |264873.34             |0.0000            |1                 |0.0043    |Low      |
|C1103251714|1                   |519523.14             |0.0000            |1                 |0.0084    |Low      |
|C877003203 |1                   |5540.45               |0.0000            |1                 |0.0001    |Low      |
+-----------+--------------------+----------------------+------------------+------------------+----------+---------+
only showing top 5 rows

>>> OUTPUT 2 SAMPLE: DAILY SUMMARY TABLE (R2) <<<
+--------+----+-----------------+-------------+-----------+
|type    |step|transaction_volume|total_amount |fraud_count|
+--------+----+-----------------+-------------+-----------+
|DEBIT   |393 |130              |6088850.81   |0          |
|DEBIT   |186 |204              |492761.48    |0          |
|TRANSFER|255 |2418             |1031491136.63|4          |
|TRANSFER|188 |3101             |1310049201.30|5          |
|CASH_IN |9   |8087             |1388800804.80|0          |
+--------+----+-----------------+-------------+-----------+
only showing top 5 rows
```

Verification in HDFS:

hdfs dfs -ls /finsight/processed/risk_scores

hdfs dfs -ls /finsight/processed/daily_summary

```
clouduser@AZ-LKM-BDH-01:~$ hdfs dfs -ls /finsight/processed/risk_scores
Found 3 items
-rw-r--r--   1 clouduser supergroup          0 2026-09-02 04:14 /finsight/processed/risk_scores/_SUCCESS
-rw-r--r--   1 clouduser supergroup    1079939 2026-09-02 04:14 /finsight/processed/risk_scores/part-00000-fbba5e8f-f086-4bae-8be5-3b877963289c-c000.snappy.parquet
-rw-r--r--   1 clouduser supergroup    1095467 2026-09-02 04:14 /finsight/processed/risk_scores/part-00001-fbba5e8f-f086-4bae-8be5-3b877963289c-c000.snappy.parquet
clouduser@AZ-LKM-BDH-01:~$ hdfs dfs -ls /finsight/processed/daily_summary
Found 3 items
-rw-r--r--   1 clouduser supergroup          0 2026-09-02 04:15 /finsight/processed/daily_summary/_SUCCESS
-rw-r--r--   1 clouduser supergroup      21891 2026-09-02 04:15 /finsight/processed/daily_summary/part-00000-4042b1ea-b016-4f32-b86c-320a49ac5b11-c000.snappy.parquet
-rw-r--r--   1 clouduser supergroup      21887 2026-09-02 04:15 /finsight/processed/daily_summary/part-00001-4042b1ea-b016-4f32-b86c-320a49ac5b11-c000.snappy.parquet
clouduser@AZ-LKM-BDH-01:~$
```

7.4) Spark Core — Customer Lifetime Value(CLV) Scoring

STEP 1: Create the HDFS Output Folder

hdfs dfs -mkdir -p /finsight/processed/clv_scores

```
clouduser@AZ-LKM-BDH-01:~$ hdfs dfs -mkdir -p /finsight/processed/clv_scores
```

```
clouduser@AZ-LKM-BDH-01:~$ hdfs dfs -ls /finsight/processed/
Found 5 items
drwxr-xr-x   - clouduser supergroup          0 2026-09-01 09:45 /finsight/processed/churn_alerts
drwxr-xr-x   - clouduser supergroup          0 2026-09-02 06:05 /finsight/processed/clv_scores
drwxr-xr-x   - clouduser supergroup          0 2026-09-02 05:25 /finsight/processed/daily_summary
drwxr-xr-x   - clouduser supergroup          0 2026-09-02 05:24 /finsight/processed/risk_scores
drwxr-xr-x   - clouduser supergroup          0 2026-09-01 04:41 /finsight/processed/streaming_metrics
clouduser@AZ-LKM-BDH-01:~$
```

STEP 2: Create the clv_scoring.py Script

nano /home/clouduser/capstone/clv_scoring.py

spark-submit /home/clouduser/capstone/clv_scoring.py

STEP 3: CLV Component Validation Table

```
+----------+----------+-----------------+------------+------------+------------+-----------+-----------+------------------+-----------------+
|min_volume|max_volume|min_frequency    |max_frequency|min_diversity|max_diversity|min_recency|max_recency|min_clv           |max_clv          |
+----------+----------+-----------------+------------+------------+------------+-----------+-----------+------------------+-----------------+
|0.0       |1.0       |0.3333333333333333|1.0         |0.2         |0.6         |0.0        |1.0        |0.13333333333333333|0.4333333333333333|
+----------+----------+-----------------+------------+------------+------------+-----------+-----------+------------------+-----------------+
```

STEP 4: Customer segment distribution

```
+-----------------+-------+
|     clv_segment|  count|
+-----------------+-------+
|          At Risk|6353300|
|Growth Potential|      7|
+-----------------+-------+
```

STEP 5: Customer CLV Ranking Table

```
+-----------+--------------------+------------------+---------------+-------------+-------------------+---------------+
|customerId |volume_score        |frequency_score   |diversity_score|recency_score|clv_score          |clv_segment    |
+-----------+--------------------+------------------+---------------+-------------+-------------------+---------------+
|C1715283297|1.0                 |0.3333333333333333|0.2            |0.0          |0.4333333333333333 |Growth Potential|
|C256397271 |0.5015916781618843  |0.6666666666666666|0.4            |0.0          |0.41714417011523197|Growth Potential|
|C545315117 |0.026885691489819334|1.0               |0.6            |0.0          |0.4080657074469458 |Growth Potential|
|C1976208114|0.005528990140110704|1.0               |0.6            |0.0          |0.40165869704203316|Growth Potential|
|C1784010646|0.00472386618488587 |1.0               |0.6            |0.0          |0.4014171598554658 |Growth Potential|
|C1832548028|0.0033929804429724043|1.0               |0.6            |0.0          |0.4010178941328917 |Growth Potential|
|C2051359467|9.661188908468411E-4|1.0               |0.6            |0.0          |0.40028983566725407|Growth Potential|
|C1741076564|0.4233555911900846  |0.6666666666666666|0.4            |0.0          |0.39367334402369203|At Risk         |
|C2127282686|0.7985621482054384  |0.3333333333333333|0.2            |0.0          |0.3729019777949648 |At Risk         |
|C385076757 |0.3278880977867181  |0.6666666666666666|0.4            |0.0          |0.3650330960026821 |At Risk         |
|C2044643633|0.7698856905863682  |0.3333333333333333|0.2            |0.0          |0.3642990405092438 |At Risk         |
|C1425667947|0.7559775080510599  |0.3333333333333333|0.2            |0.0          |0.36012658574865125|At Risk         |
|C1584456031|0.7500343855507063  |0.3333333333333333|0.2            |0.0          |0.3583436489985452 |At Risk         |
|C1162922333|0.06827166432070253 |0.3333333333333333|0.2            |1.0          |0.3538148326295441 |At Risk         |
|C1529008245|0.06827166432070253 |0.3333333333333333|0.2            |1.0          |0.3538148326295441 |At Risk         |
|C1902386530|0.008261220422122139|1.0               |0.4            |0.0          |0.35247836612663663|At Risk         |
|C811810230 |0.7301680356534812  |0.3333333333333333|0.2            |0.0          |0.35238374402937767|At Risk         |
|C1530544995|0.005306207459581136|1.0               |0.4            |0.0          |0.3515918622378743 |At Risk         |
|C1999539787|0.0031429864915080213|1.0               |0.4            |0.0          |0.3509428959474524 |At Risk         |
|C1677795071|0.0026473472040082265|1.0               |0.4            |0.0          |0.35079420416120244|At Risk         |
+-----------+--------------------+------------------+---------------+-------------+-------------------+---------------+
only showing top 20 rows
```

STEP 6 : Final Validation Box & Hive Table Registration

```
============================================================
Hive external table registered: finsight.customer_clv
============================================================

Hive table row count:
Hive customer count: 6353307

============================================================
FINAL 7.4 VALIDATION
============================================================
Input transactions : 6362620
Output customers   : 6353307
Latest step        : 743
HDFS output        : /finsight/processed/clv_scores
Hive table         : finsight.customer_clv
Spark appName      : FinSight-CLV-Scoring
FinSight 7.4 CLV JOB COMPLETED SUCCESSFULLY.
```

7.5) Spark SQL — Compliance Aggregation

STEP 1: Create HDFS Output Directories

hdfs dfs -mkdir -p /finsight/processed/customer_fraud_summary

hdfs dfs -mkdir -p /finsight/processed/weekly_compliance_summary

```
clouduser@AZ-LKM-BDH-01:~$ hdfs dfs -mkdir -p /finsight/processed/customer_fraud_summary
clouduser@AZ-LKM-BDH-01:~$ hdfs dfs -mkdir -p /finsight/processed/weekly_compliance_summary
clouduser@AZ-LKM-BDH-01:~$ hdfs dfs -ls /finsight/processed/
Found 7 items
drwxr-xr-x   - clouduser supergroup          0 2026-09-01 09:45 /finsight/processed/churn_alerts
drwxr-xr-x   - clouduser supergroup          0 2026-09-02 06:38 /finsight/processed/clv_scores
drwxr-xr-x   - clouduser supergroup          0 2026-09-02 07:45 /finsight/processed/customer_fraud_summary
drwxr-xr-x   - clouduser supergroup          0 2026-09-02 05:25 /finsight/processed/daily_summary
drwxr-xr-x   - clouduser supergroup          0 2026-09-02 05:24 /finsight/processed/risk_scores
drwxr-xr-x   - clouduser supergroup          0 2026-09-01 04:41 /finsight/processed/streaming_metrics
drwxr-xr-x   - clouduser supergroup          0 2026-09-02 07:46 /finsight/processed/weekly_compliance_summary
clouduser@AZ-LKM-BDH-01:~$
```

STEP 2: Create the spark_sql_jobs.py Script

nano /home/clouduser/capstone/spark_sql_jobs.py

STEP 3: Run the 2 Modes

For Compliance Summary:

spark-submit /home/clouduser/capstone/spark_sql_jobs.py --mode compliance

```
============================================================
                EXECUTION SUMMARY
============================================================
Status          : SUCCESS
Mode            : compliance
Source records  : 6,362,620
Output rows     : 5
Execution time  : 45.08 seconds
============================================================

# ==================== COMPLIANCE SUMMARY ====================
+--------+---------+--------------------+-----------+--------------+
|type    |txn_count|total_amount        |fraud_count|fraud_rate_pct|
+--------+---------+--------------------+-----------+--------------+
|CASH_IN |37321    |6.219063114970016E9 |0          |0.00          |
|CASH_OUT|52566    |9.992564793579872E9 |927        |1.76          |
|DEBIT   |1328     |4083965.1600000034  |0          |0.00          |
|PAYMENT |55537    |6.261208269400014E8 |0          |0.00          |
|TRANSFER|15551    |1.1210895257019917E10|927        |5.96          |
+--------+---------+--------------------+-----------+--------------+

============================================================
[COMPLETE] Task 7.5 finished successfully
============================================================
```

For Customer Fraud Summary:

spark-submit /home/clouduser/capstone/spark_sql_jobs.py --mode customer_summary

```
============================================================
                EXECUTION SUMMARY
============================================================
Status          : SUCCESS
Mode            : customer_summary
Source records  : 6,362,620
Output rows     : 6,353,307
Execution time  : 90.71 seconds
============================================================

# ================ CUSTOMER FRAUD SUMMARY ==================
+-----------+----------------------+------------+--------------------+--------------+
|customerId |total_transaction_count|total_amount|confirmed_fraud_count|fraud_rate_pct|
+-----------+----------------------+------------+--------------------+--------------+
|C1000331499|1                     |2016790.84  |1                   |100.00        |
|C1000036340|1                     |253648.68   |1                   |100.00        |
|C1005866728|1                     |145379.14   |1                   |100.00        |
|C1005007866|1                     |295205.08   |1                   |100.00        |
|C1000484178|1                     |3018810.85  |1                   |100.00        |
|C1000086512|1                     |33676.59    |1                   |100.00        |
|C1000937208|1                     |1534985.07  |1                   |100.00        |
|C1000513158|1                     |40388.58    |1                   |100.00        |
|C1001875185|1                     |3288377.47  |1                   |100.00        |
|C1001502110|1                     |484493.06   |1                   |100.00        |
+-----------+----------------------+------------+--------------------+--------------+
only showing top 10 rows

============================================================
[COMPLETE] Task 7.5 finished successfully
============================================================
```

7.6) Spark SQL — Account Dormancy Report

STEP 1: Create HDFS Output & Export Folders

hdfs dfs -mkdir -p /finsight/processed/dormancy_report

hdfs dfs -mkdir -p /finsight/exports/dormancy_report

```
clouduser@AZ-LKM-BDH-01:~$ hdfs dfs -mkdir -p /finsight/processed/dormancy_report
clouduser@AZ-LKM-BDH-01:~$ hdfs dfs -mkdir -p /finsight/exports/dormancy_report
```

STEP 2: Update spark_sql_jobs.py

STEP 3: Execute Task

spark-submit /home/clouduser/capstone/spark_sql_jobs.py --mode dormancy

```
======================================================================
                  EXECUTION SUMMARY
======================================================================
Status          : SUCCESS
Mode            : dormancy
Source records  : 6,362,620
Output rows     : 0
Execution time  : 109.21 seconds
======================================================================

# ================= ACCOUNT DORMANCY REPORT ==================
+----------+----------------+--------------+-----------------------+-----------------+
|customerId|last_active_step|steps_inactive|prior_transaction_count|dormancy_severity|
+----------+----------------+--------------+-----------------------+-----------------+
+----------+----------------+--------------+-----------------------+-----------------+

======================================================================
[COMPLETE] Mode 'dormancy' finished successfully
======================================================================
```

DAY 3 – Database Configuration

8.1) Hive — Data Warehouse Layer

STEP 1: Create Database

```
hive> CREATE DATABASE IF NOT EXISTS finsight;
OK
Time taken: 0.668 seconds
hive> USE finsight;
OK
Time taken: 0.055 seconds
```

STEP 2 : Create External Table

```
hive> DROP TABLE IF EXISTS finsight.transactions;
OK
Time taken: 0.177 seconds
hive> CREATE EXTERNAL TABLE IF NOT EXISTS finsight.transactions (
    >     type STRING,
    >     amount DOUBLE,
    >     nameOrig STRING,
    >     oldbalanceOrg DOUBLE,
    >     newbalanceOrig DOUBLE,
    >     nameDest STRING,
    >     oldbalanceDest DOUBLE,
    >     newbalanceDest DOUBLE,
    >     isFraud INT,
    >     isFlaggedFraud INT
    > )
    > PARTITIONED BY (step INT)
    > STORED AS PARQUET
    > LOCATION '/finsight/raw/transactions/txn-raw/';
OK
Time taken: 0.162 seconds
hive> MSCK REPAIR TABLE finsight.transactions;
OK
```

STEP 3: Create the Fraud View (Requirement R1)

```
hive>
    > DROP VIEW IF EXISTS finsight.vw_fraud_transactions;
OK
Time taken: 0.177 seconds
hive> CREATE VIEW finsight.vw_fraud_transactions AS
    > SELECT step, type, amount, nameOrig, nameDest, isFraud
    > FROM finsight.transactions
    > WHERE isFraud = 1;
OK
Time taken: 0.491 seconds
```

STEP 4: Compute Table Statistics (Requirement R2)

```
hive> ANALYZE TABLE finsight.transactions COMPUTE STATISTICS;
Table finsight.transactions stats: [numFiles=8471, numRows=6361000, totalSize=317020447, rawDataSize=387039084]
OK
Time taken: 126.963 seconds
```

STEP 5: Verify Your Tables & Sample Data

```
hive> SELECT * FROM finsight.vw_fraud_transactions LIMIT 5;
OK
1       CASH_OUT        2806.0  C2101527076     C1007251739     1
1       TRANSFER        20128.0 C137533655      C1848415041     1
1       TRANSFER        2806.0  C1420196421     C972765878      1
1       CASH_OUT        416001.33       C749981943      C667346055      1
1       TRANSFER        1277212.77      C1334405552     C431687661      1
Time taken: 0.45 seconds, Fetched: 5 row(s)
```

8.2) Hive — Transaction Summary Mart

STEP 1: Create the finsight.txn_summary_mart Table

```
hive> USE finsight;
OK
Time taken: 0.807 seconds
hive> DROP TABLE IF EXISTS finsight.txn_summary_mart;
OK
Time taken: 0.134 seconds
hive> CREATE TABLE IF NOT EXISTS finsight.txn_summary_mart (
    >     customerId STRING,
    >     step INT,
    >     txn_count BIGINT,
    >     total_amount DOUBLE,
    >     avg_amount DOUBLE,
    >     max_amount DOUBLE,
    >     fraud_count BIGINT,
    >     txn_types STRING,
    >     last_balance DOUBLE
    > )
    > STORED AS PARQUET;
OK
Time taken: 0.757 seconds
hive>
```

STEP 2: Fill Table With Data (Calculate Customer Totals)

```
hive> INSERT OVERWRITE TABLE finsight.txn_summary_mart
    > SELECT
    >     nameOrig AS customerId,
    >     step,
    >     COUNT(*) AS txn_count,
    >     ROUND(SUM(amount), 2) AS total_amount,
    >     ROUND(AVG(amount), 2) AS avg_amount,
    >     ROUND(MAX(amount), 2) AS max_amount,
    >     SUM(CASE WHEN isFraud = 1 OR isFraud = true THEN 1 ELSE 0 END) AS fraud_count,
    >     concat_ws(',', collect_set(type)) AS txn_types,
    >     MAX(newbalanceOrig) AS last_balance
    > FROM finsight.transactions
    > GROUP BY nameOrig, step;
Query ID = clouduser_20260903095555_f9002e38-1664-49ca-97d8-53f81cb160aa
Total jobs = 2

Hadoop job information for Stage-3: number of mappers: 1; number of reducers: 1
2026-09-03 10:09:01,156 Stage-3 map = 0%,  reduce = 0%
2026-09-03 10:09:07,412 Stage-3 map = 100%,  reduce = 0%, Cumulative CPU 1.68 sec
2026-09-03 10:09:13,659 Stage-3 map = 100%,  reduce = 100%, Cumulative CPU 3.74 sec
MapReduce Total cumulative CPU time: 3 seconds 740 msec
Ended Job = job_1788407756806_0002
MapReduce Jobs Launched:
Stage-Stage-1: Map: 2  Reduce: 2   Cumulative CPU: 836.52 sec   HDFS Read: 165372060 HDFS Write: 10206 SUCCESS
Stage-Stage-3: Map: 1  Reduce: 1   Cumulative CPU: 3.74 sec   HDFS Read: 27893 HDFS Write: 6597 SUCCESS
Total MapReduce CPU Time Spent: 14 minutes 0 seconds 260 msec
OK
Time taken: 800.933 seconds
hive>
```

STEP 3: Compute Column Statistics (Requirement R1)

```
hive> ANALYZE TABLE finsight.txn_summary_mart COMPUTE STATISTICS FOR COLUMNS;
Query ID = clouduser_20260903101306_f2d2a4ee-4ec3-4cd4-95b9-ab157ab9559e
Total jobs = 1
Launching Job 1 out of 1

MapReduce Jobs Launched:
Stage-Stage-0: Map: 1  Reduce: 1   Cumulative CPU: 24.51 sec   HDFS Read: 19678 HDFS Write: 6689 SUCCESS
Total MapReduce CPU Time Spent: 24 seconds 510 msec
OK
Time taken: 40.567 seconds
hive>
```

STEP 4: Verify the Table Data

```
hive> SELECT * FROM finsight.txn_summary_mart
    > WHERE fraud_count > 0
    > LIMIT 5;
OK
txn_summary_mart.customerid     txn_summary_mart.step   txn_summary_mart.txn_count      txn_summary_mart.total_amount   txn_summary_mart.avg_amount     txn_summary_mart.max_amount     txn_summary_mart.fraud_coun
t       txn_summary_mart.txn_types      txn_summary_mart.last_balance
C1000036340     655     1       253648.68       253648.68       253648.68       1       TRANSFER        0.0
C1000086512     95      1       33676.59        33676.59        33676.59        1       CASH_OUT        0.0
C1000331499     262     1       2016790.84      2016790.84      2016790.84      1       TRANSFER        0.0
C1000484178     504     1       3018810.85      3018810.85      3018810.85      1       CASH_OUT        0.0
C1000937208     82      1       1534985.07      1534985.07      1534985.07      1       TRANSFER        0.0
Time taken: 0.488 seconds, Fetched: 5 row(s)
hive>
```

8.3) MongoDB — Customer Document Store

STEP 1: Import the 10,000 Customer JSON Documents

mongoimport --db finsight --collection customers --drop --file "/home/clouduser/Downloads/Bigdata-Dataset/src-data/noveacrest_customers.json" –jsonArray

```
clouduser@AZ-LKM-BDH-01:~$ mongoimport --db finsight --collection customers --drop --file "/home/clouduser/Downloads/Bigdata-Dataset/src-data/noveacrest_customers.json" --jsonArray
2026-09-04T05:05:04.917+0000    connected to: mongodb://localhost/
2026-09-04T05:05:04.917+0000    dropping: finsight.customers
2026-09-04T05:05:05.713+0000    10000 document(s) imported successfully. 0 document(s) failed to import.
clouduser@AZ-LKM-BDH-01:~$
```

STEP 2: Open the MongoDB Shell

mongosh

STEP 3: Create the Compound Index (Requirement R1):

use finsight

db.customers.createIndex({ customerId: 1, segment: 1 })

```
test> use finsight
switched to db finsight
finsight> db.customers.createIndex({ customerId: 1, segment: 1 })
customerId_1_segment_1
finsight>
```

STEP 4: Run the Aggregation Pipeline Query (Requirement R2):

```
finsight> db.customers.aggregate([
...     {
...       $group: {
...         _id: "$segment",
...         count: { $sum: 1 }
...       }
...     },
...     {
...       $sort: { count: -1 }
...     }
... ])
[
  { _id: 'Standard', count: 4497 },
  { _id: 'Basic', count: 2464 },
  { _id: 'Premium', count: 1528 },
  { _id: 'Student', count: 983 },
  { _id: 'Private Banking', count: 528 }
]
finsight>
```

STEP 5: Verify 1 Sample Customer Document

```
finsight> db.customers.findOne()
{
  _id: ObjectId('6a9a518155f95d8915904b46'),
  customerId: 'C2801823908',
  name: 'Anna Johnson',
  age: 31,
  country: 'UK',
  segment: 'Standard',
  annualIncome: 70753,
  products: [ 'fixed_deposit', 'car_loan' ],
  productCount: 2,
  kyc_status: 'verified',
  risk_score: 0.1335,
  churn_probability: 0.3147,
  composite_risk_score: 0.206,
  account_opened: '2019-04-06',
  last_transaction_date: '2023-04-10',
  preferred_channel: 'branch',
  email_verified: true,
  is_active: true
}
finsight>
```

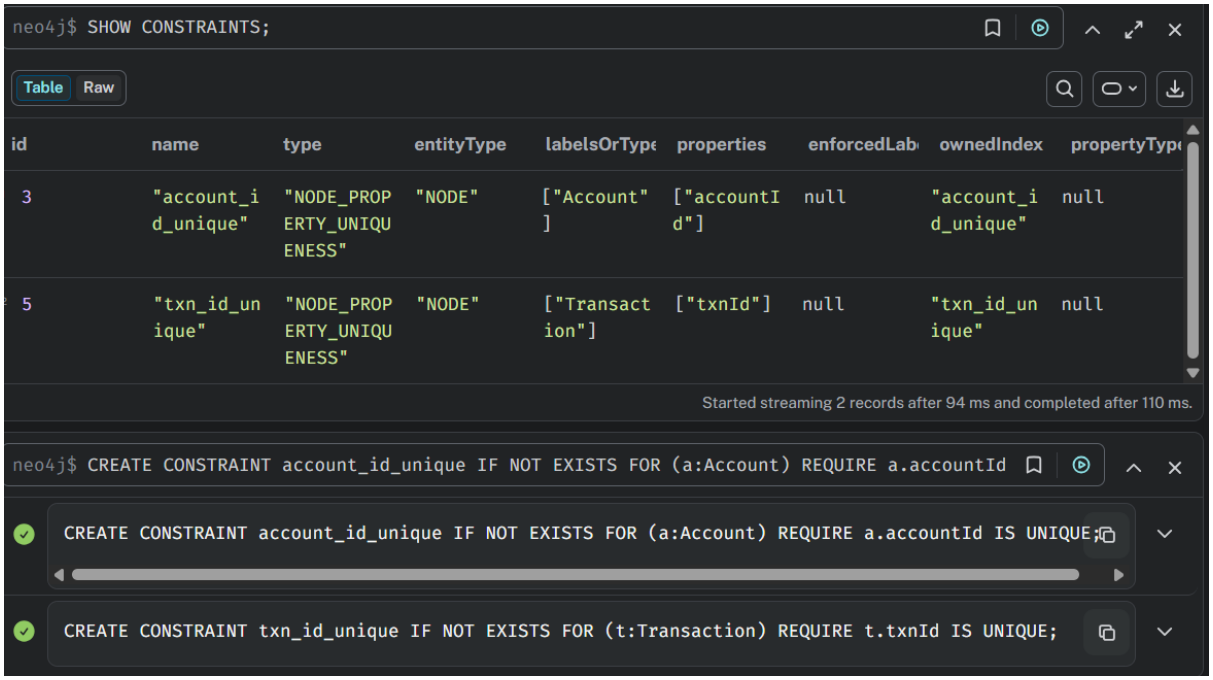
8.4) Neo4j — Fraud Knowledge Graph

STEP 1: Create Constraints (For Speed & Uniqueness)

CREATE CONSTRAINT account_id_unique IF NOT EXISTS FOR (a:Account) REQUIRE a.accountId IS UNIQUE;

CREATE CONSTRAINT txn_id_unique IF NOT EXISTS FOR (t:Transaction) REQUIRE t.txnId IS UNIQUE;

SHOW CONSTRAINTS;



STEP 2: Load Account Nodes & Verify

LOAD CSV WITH HEADERS FROM 'file:///neo4j_accounts_nodes.csv' AS row

MERGE (a:Account {accountId: row['accountId:ID']})

SET a.accountType = row.accountType;

Verify:

MATCH (a:Account) RETURN count(a) AS total_account_nodes;



STEP 3: Load Transaction Nodes & Verify

LOAD CSV WITH HEADERS FROM 'file:///neo4j_transaction_nodes.csv' AS row

MERGE (t:Transaction {txnId: row['txnId:ID']})

SET t.step = toInteger(row['step:int']),

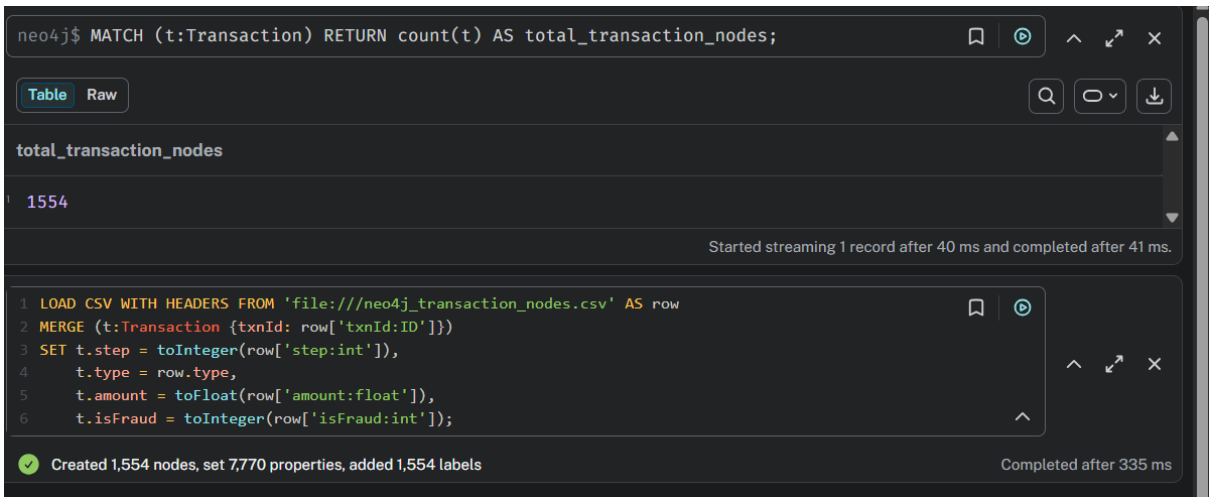
t.type = row.type,

t.amount = toFloat(row['amount:float']),

t.isFraud = toInteger(row['isFraud:int']);

Verify :

MATCH (t:Transaction) RETURN count(t) AS total_transaction_nodes;



STEP 4: Load :SENT Relationships & Verify

```
LOAD CSV WITH HEADERS FROM 'file:///neo4j_sent_rels.csv' AS row
MATCH (a:Account {accountId: row[':START_ID']})
MATCH (t:Transaction {txnId: row[':END_ID']})
MERGE (a)-[r:SENT]->(t)
SET r.amount = toFloat(row['amount:float']),
  r.step = toInteger(row['step:int']),
 r.transactionType = row.transactionType;
```

Verify :

```
MATCH ()-[r:SENT]->() RETURN count(r) AS total_sent_relationships;
```

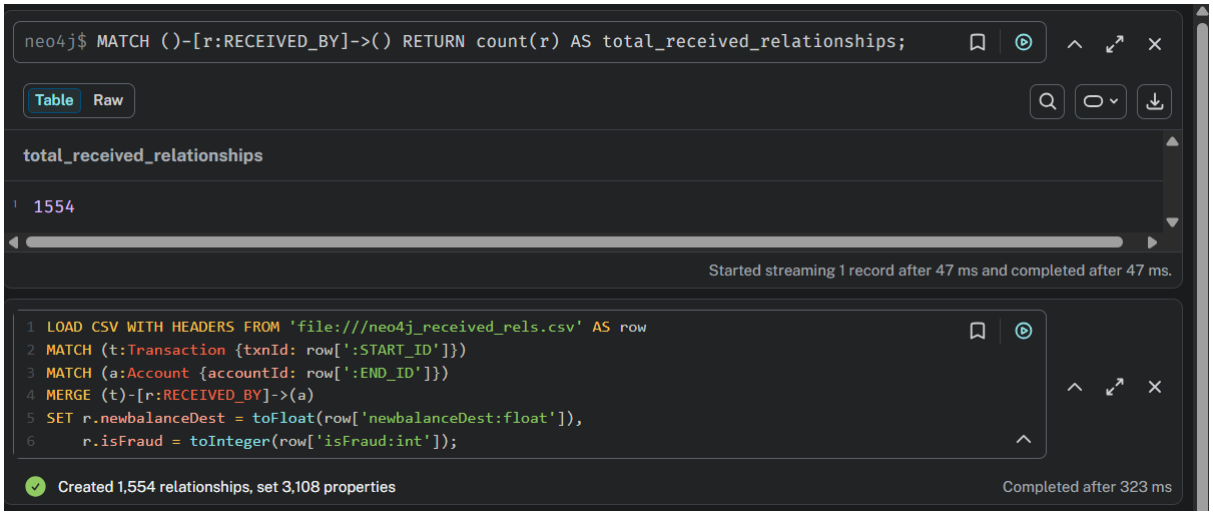


STEP 5: Load :RECEIVED_BY Relationships

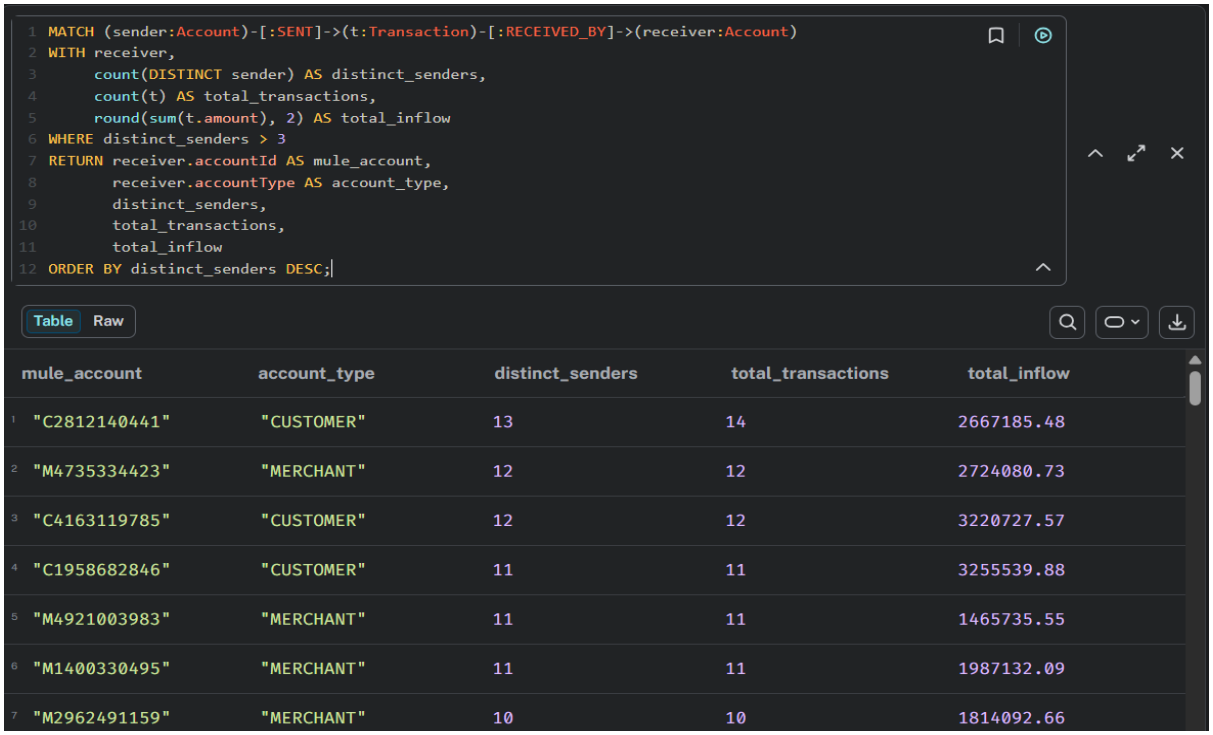
```
LOAD CSV WITH HEADERS FROM 'file:///neo4j_received_rels.csv' AS row
MATCH (t:Transaction {txnId: row[':START_ID']})
MATCH (a:Account {accountId: row[':END_ID']})
MERGE (t)-[r:RECEIVED_BY]->(a)
SET r.newbalanceDest = toFloat(row['newbalanceDest:float']),
  r.isFraud = toInteger(row['isFraud:int']);
```

Verify:

MATCH ()-[r:RECEIVED_BY]->() RETURN count(r) AS total_received_relationships;



STEP 6: Execute the Money Mule Detection Cypher Query



Graph Visualization:

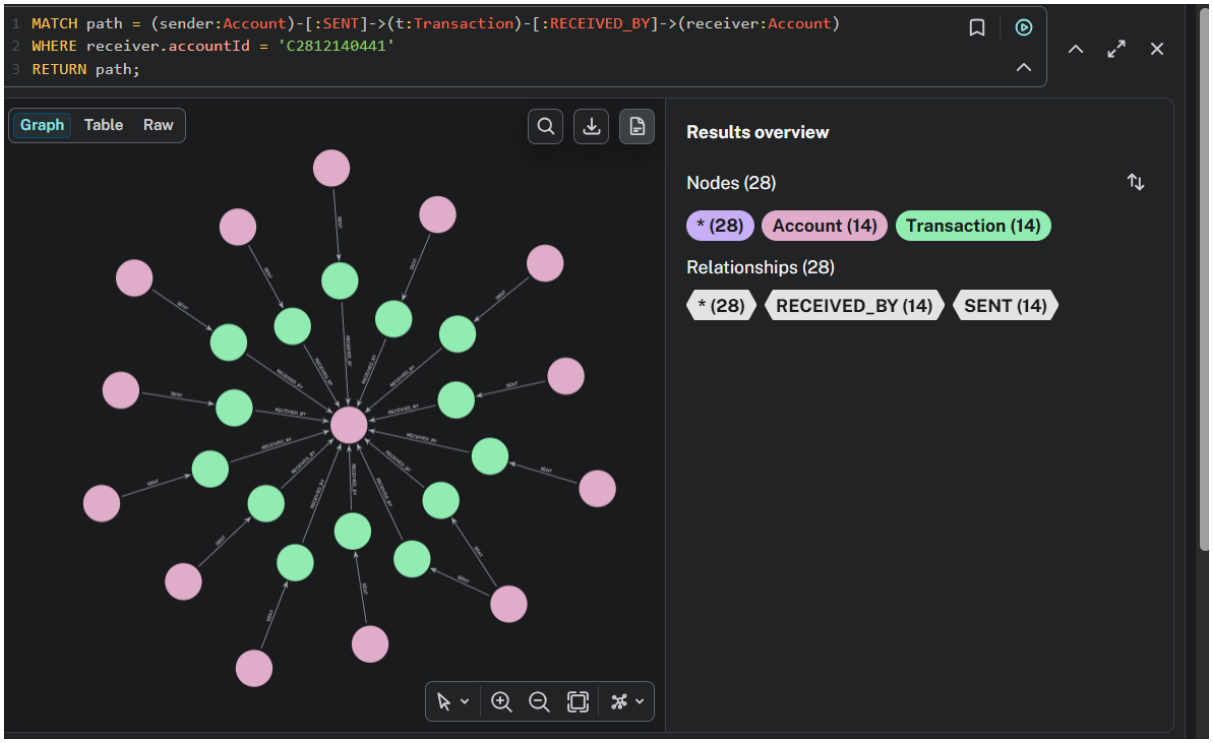


Table Visualization:



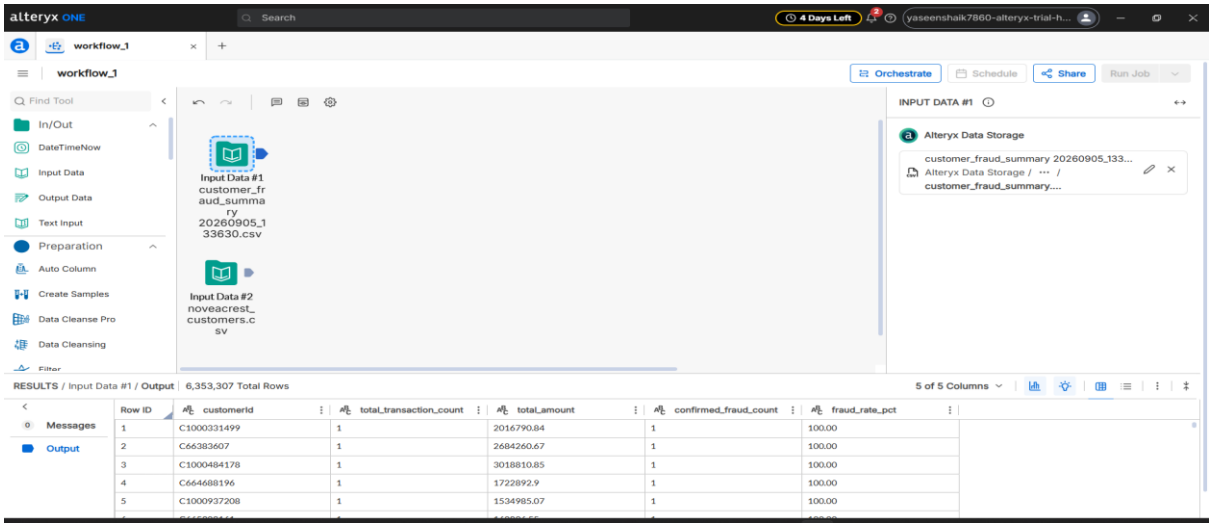
DAY 4 – Alteryx – Data Blending and Feature Engineering

9.1) Workflow 1 — Customer Risk Blend

Step 1: Load Input Data

Input A: Loaded customer_fraud_summary.csv

Input B: Loaded noveacrest_customers.csv

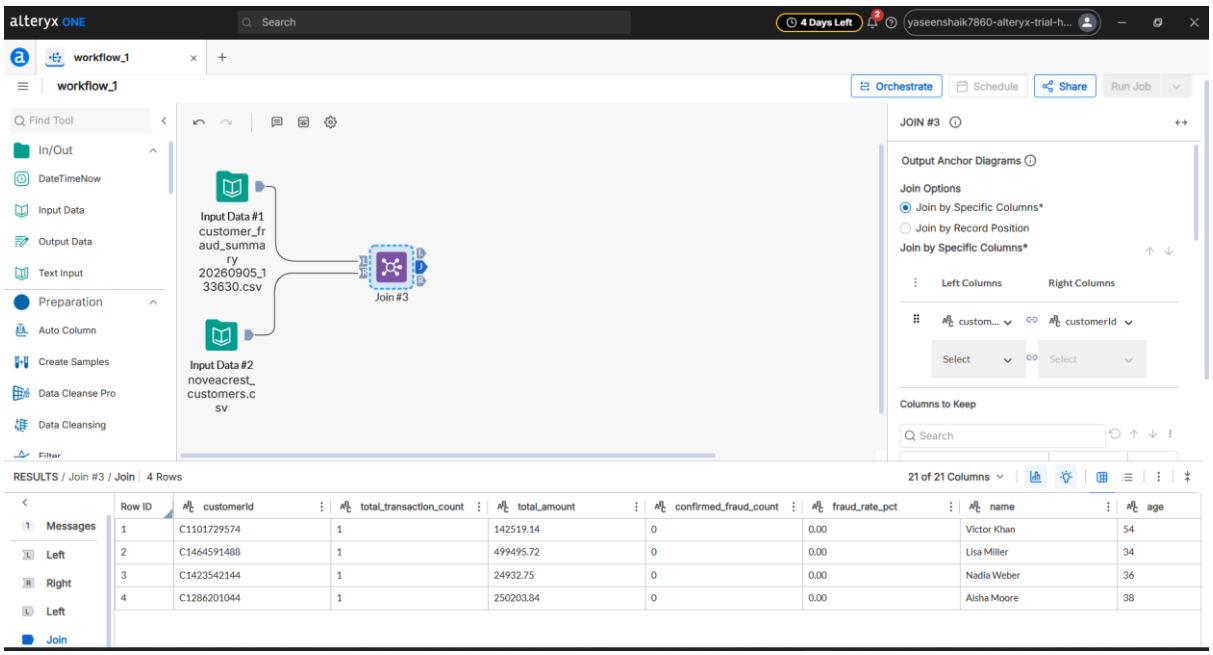


Step 2: Join Datasets

Added a Join Tool and connected Input A to Left (L) and Input B to Right (R).

Joined both datasets on customerId = customerId.

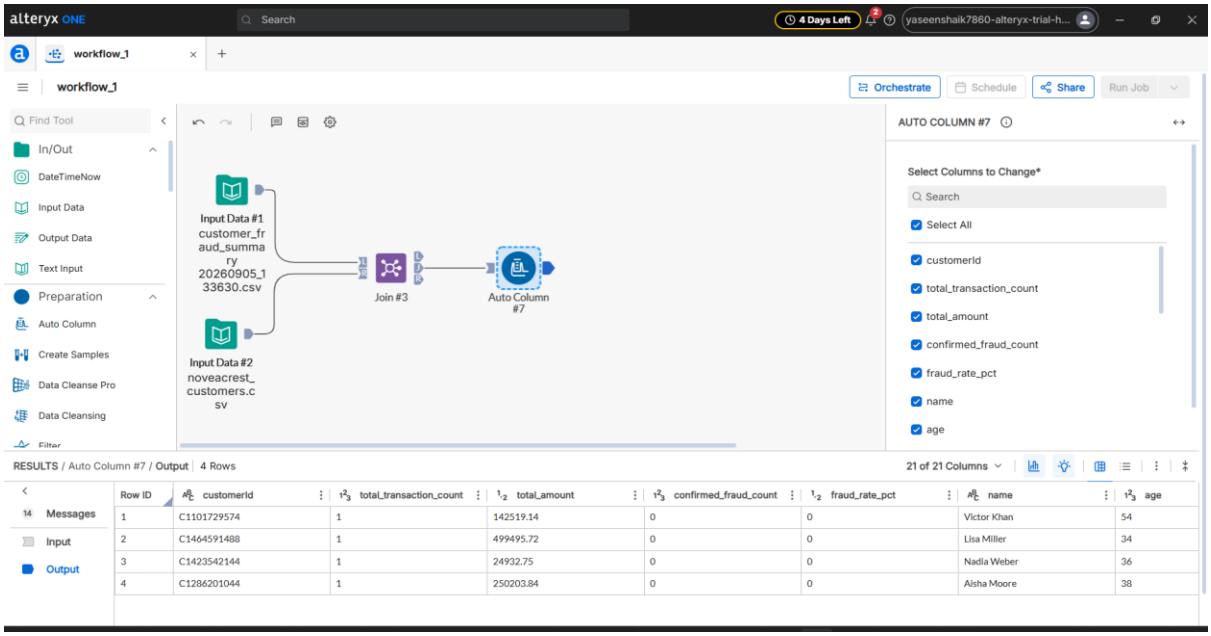
Uncheck the duplicate Right_customerId in the column list below to keep it clean.



Step 3: Auto-Detect Data Types (Auto Column Tool)

Connected an Auto Column Tool to the Join (J) output.

Automatically converted Data types

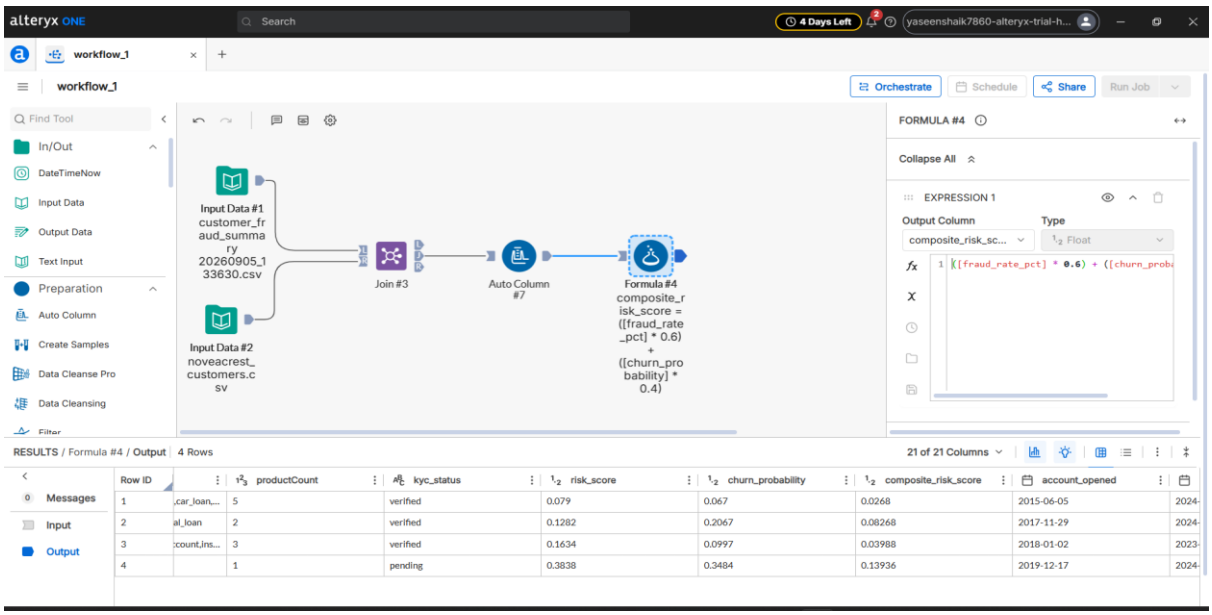


Step 4: Calculate Composite Risk Score

Connected a Formula Tool to the middle Join output (J).

Selected the existing column composite_risk_score.

Applied the formula: composite_risk_score = (fraud_rate_pct × 0.6) + (churn_probability × 0.4)

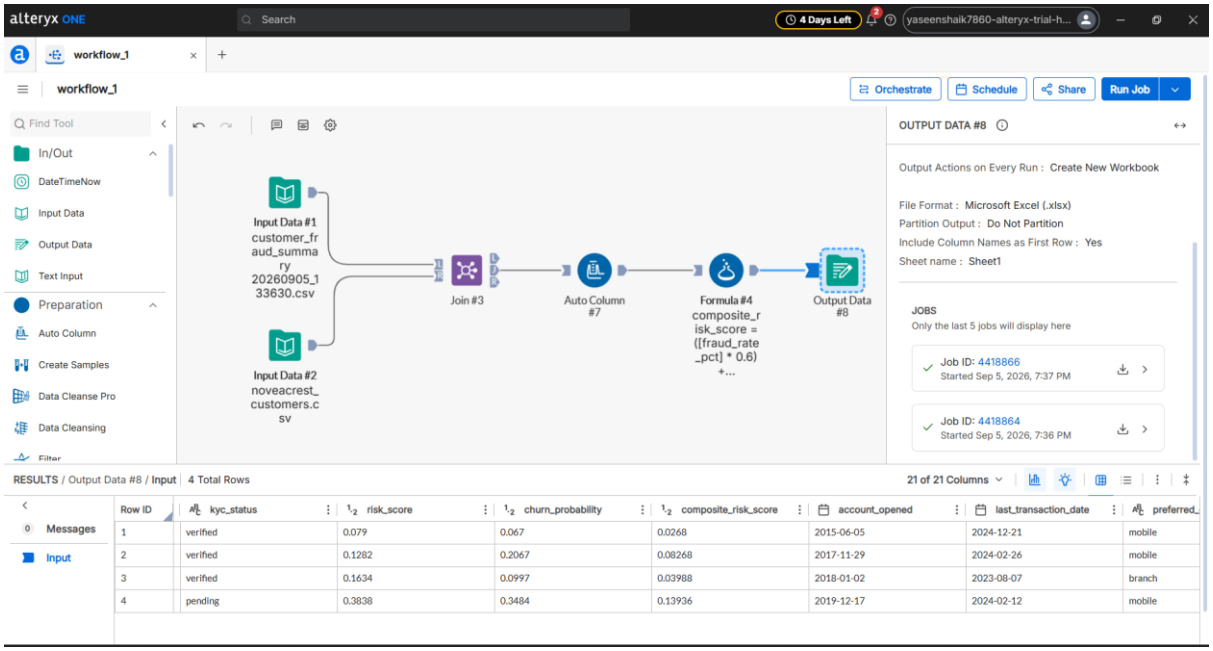


Step 5: Final Workflow 1

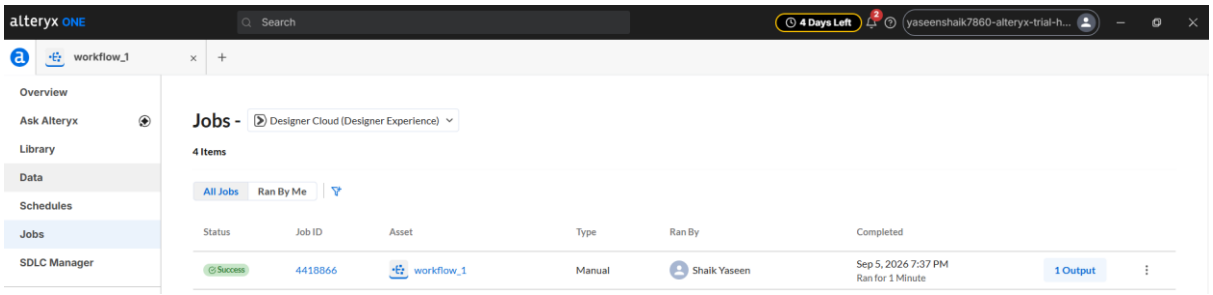
Added an Output Data Tool connected to the Formula tool.

Exported the final blended feature store as customer_risk_blend.xlsx

Clicked the Run Job button to execute the entire pipeline.



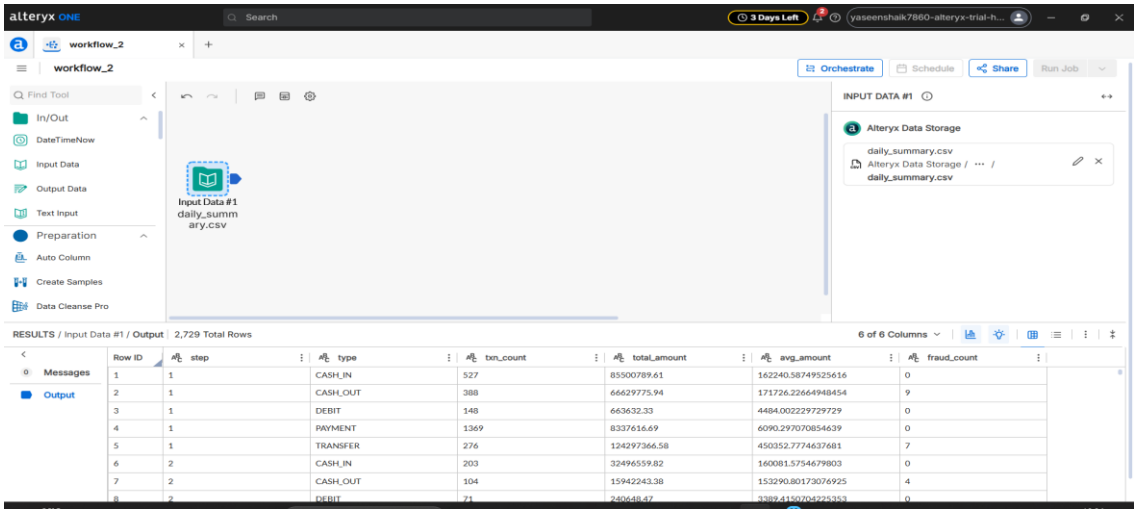
Verification of Job:



9.2) Workflow 2 — Transaction Summary

Step 1: Load Input Data

Draged an Input Data Tool to load daily_summary.csv

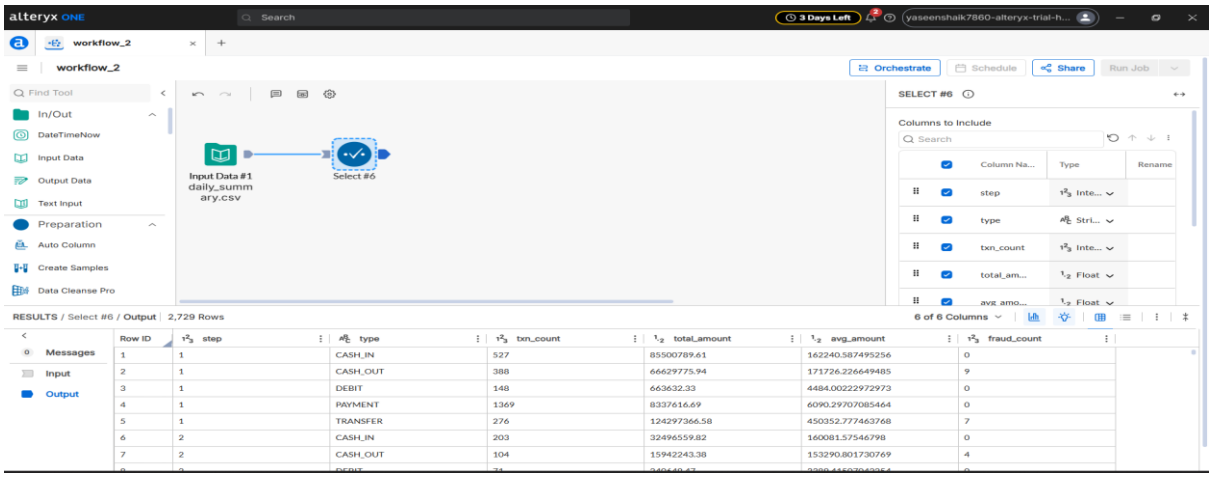


Step 2: Set Data Types (Select Tool)

Added a Select Tool connected to Input Data.

Converted avg_amount and total_amount to numeric **Float** data types for mathematical operations.

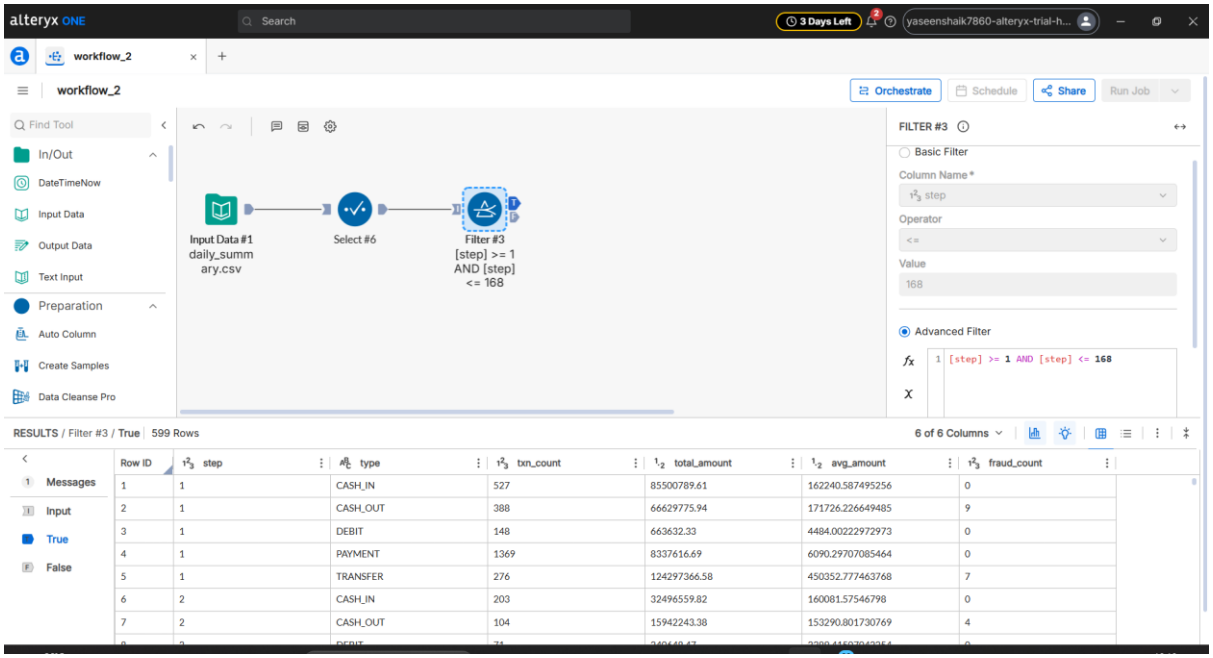
Converted step , txn_count , fraud_count to Integer.



Step 3: Filter First 7 Days (Filter Tool)

Added a Filter Tool connected to the Select tool.

Applied the condition in Advanced Filter: **[step] >= 1 AND [step] <= 168** to keep the first 7 simulation days (168 hours).



Step 4: Summarize Data (Summarize Tool)

Connected a **Summarize Tool** to the **True (T)** output of the Filter tool.

Added Action : type ⟶ Group By

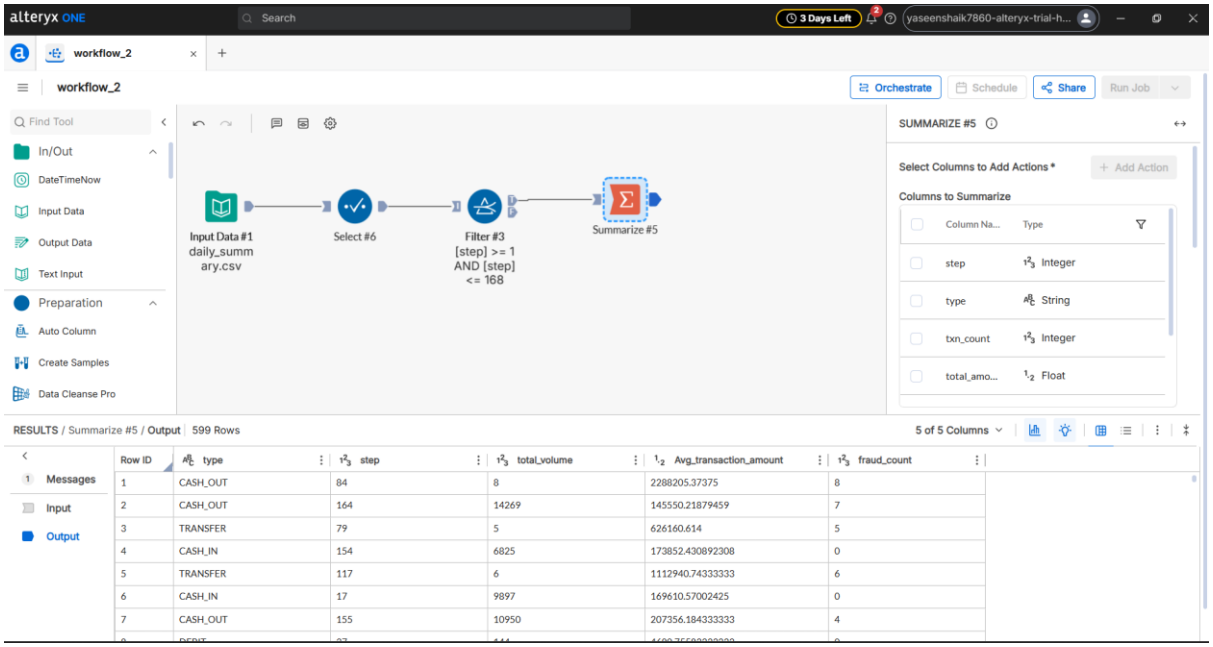
Added Action : step ⟶ Group By

Added Action : txn_count ⟶ Sum (Renamed to total_volume)

Added Action : avg_amount ⟶ Average (Renamed to Avg_Transaction_amount)

Added Action : fraud_count ⟶ Sum (Renamed to fraud_count)

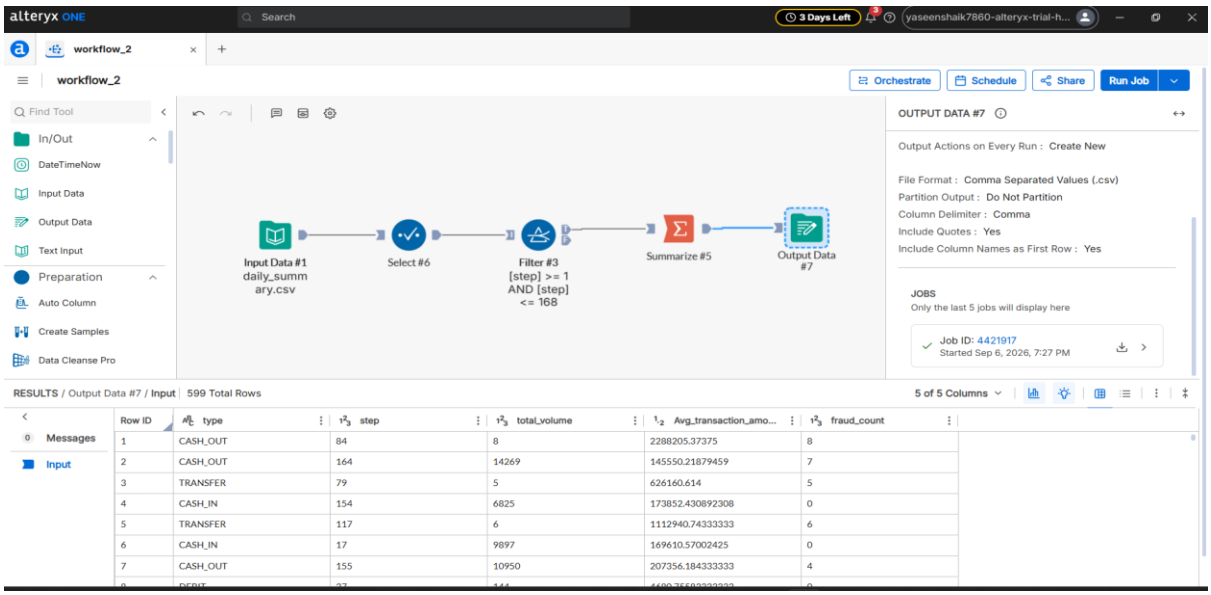
Generated 599 summarized rows.



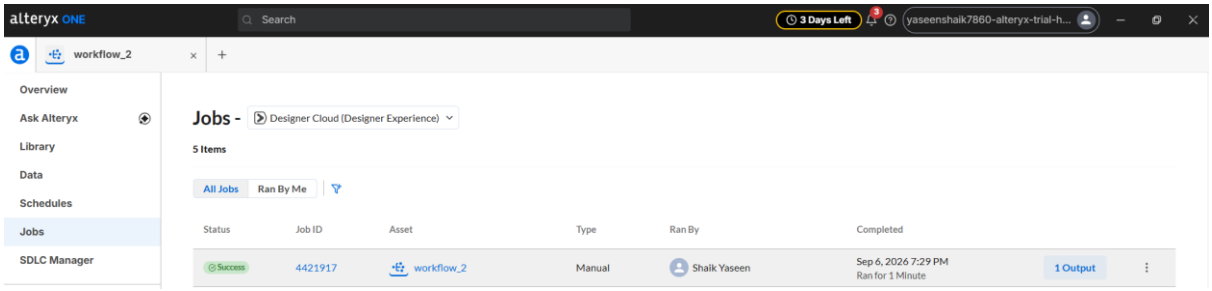
Step 5: Final Workflow

Added an Output Data Tool connected to the Summarize tool.

Exported the final summarized dataset as transaction_summary.csv



Verification of Job:



DAY 5 – Power BI Dashboards

10. Power BI Dashboards

Dashboard Page 1 — Fraud Alert Board

Step 1: Header Banner & Status Badge

Created the top header banner with the title **FinSight — Fraud AlertBoard**.

Applied background colour

Step 2: Created DAX Measures & 4 KPI Cards

1) **Created DAX Measures**: Created DAX measures to calculate the 4 KPI numbers and their green subtitle text from the streaming dataset.
2) **Applied Container & Colour Formatting**: Formatted all 4 cards with dark container backgrounds, rounded borders, coloured callout values, and green subtitle text.

CREATING 4 KPI CARDS:

1) Card 1 (Total Flagged)
2) Card 2 (Fraud Rate Pct)
3) Card 3 (Total Flagged Value)
4) Card 4 (False Positive Rate)

Step 3: Created Visual 1 — Daily Fraud Volume by Type (Line Chart)

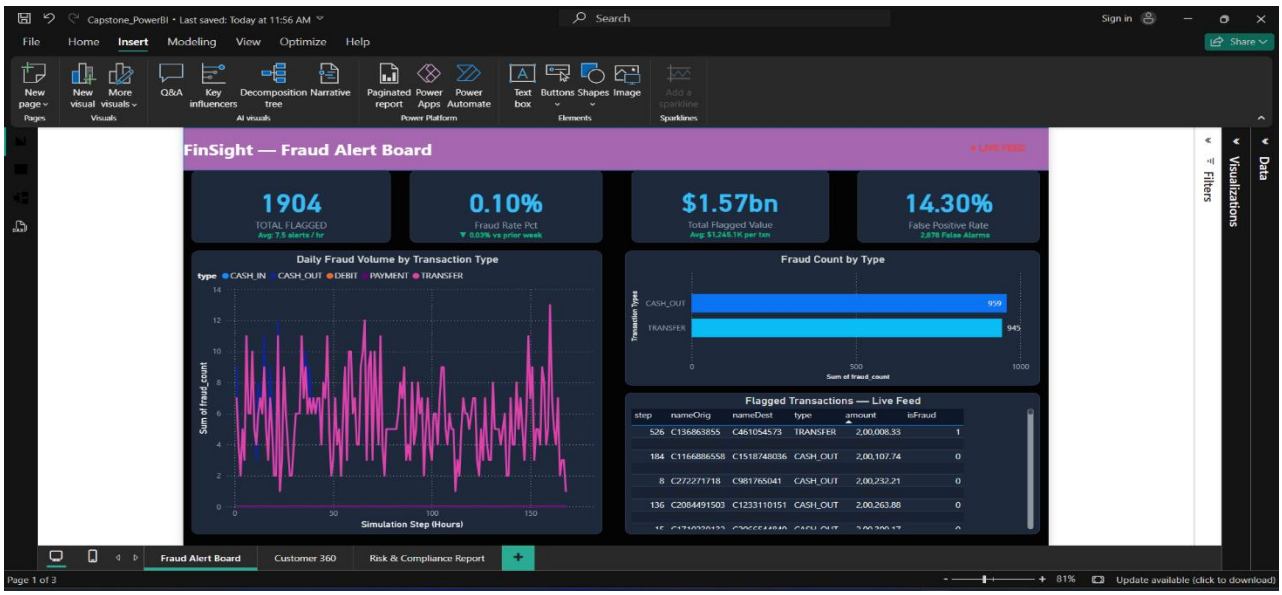
1) Placed a Line Chart on theleft.
2) Set **X-axis** to step (simulation timeline), **Y-axis** to fraud_count, and **Legend** to type
3) Applied colour formatting to lines and styled the chart background.
4) Shows how fraud counts change over time across different transaction types.

Step 4: Created Visual 2 — Fraud Count by Type (Bar Chart)

1) Placed a Clustered Bar Chart on the top right.
2) Set **Y-axis** to Transaction Types and **X-axis** to Sum of fraud_count.
3) Applied colour formatting to the bars and turned on data labels
4) Shows that fraud is concentrated in CASH_OUT (959) and TRANSFER (945), with zero fraud in other categories.

Step 5: Created Visual 3 — Flagged Transactions (Table)

1) Selected a Table visual.
2) Added columns: step, nameOrig, nameDest, type, amount, and isFraud.
3) Applied colour formatting to table rows, header, and gridlines.
4) Displays Table.



Dashboard Page 2 — FinSight — Customer 360 (Portfolio View)

Step 1: Header Banner & Status Badge

1) Created the top header banner
2) Added the **5 SEGMENTS** badge on the top right to indicate the 5 customer portfolio tiers.
3) Applied Background colour.

Step 2: Created the 4 KPI Cards with the help of DAX measures

Created DAX Measures: Built dynamic DAX measures for Total Customers, Average Risk Score, Average Churn Probability, and Average Products Held, along with subtitle benchmark measures.

Applied Container & Color Formatting: Formatted all 4 cards with dark container backgrounds, rounded borders, coloured callout values.

1) Card 1 (Total Customers)

2) Card 2 (Avg Risk Score)
3) Card 3 (Avg Churn Prob)
4) Card 4 (Avg Products Held)

Step 3: Created Visual 1 — Risk Score vs Churn Probability (Scatter Plot)

1) Selected a Scatter Plot.
2) Set X-axis to Risk Score, Y-axis to Churn Probability, and Legend to Segment.
3) Coloured the 5 segments
4) Helps relationship managers identify high-risk, high-churn customer

Step 4: Created Visual 2 — Customer Count by Segment (Bar Chart)

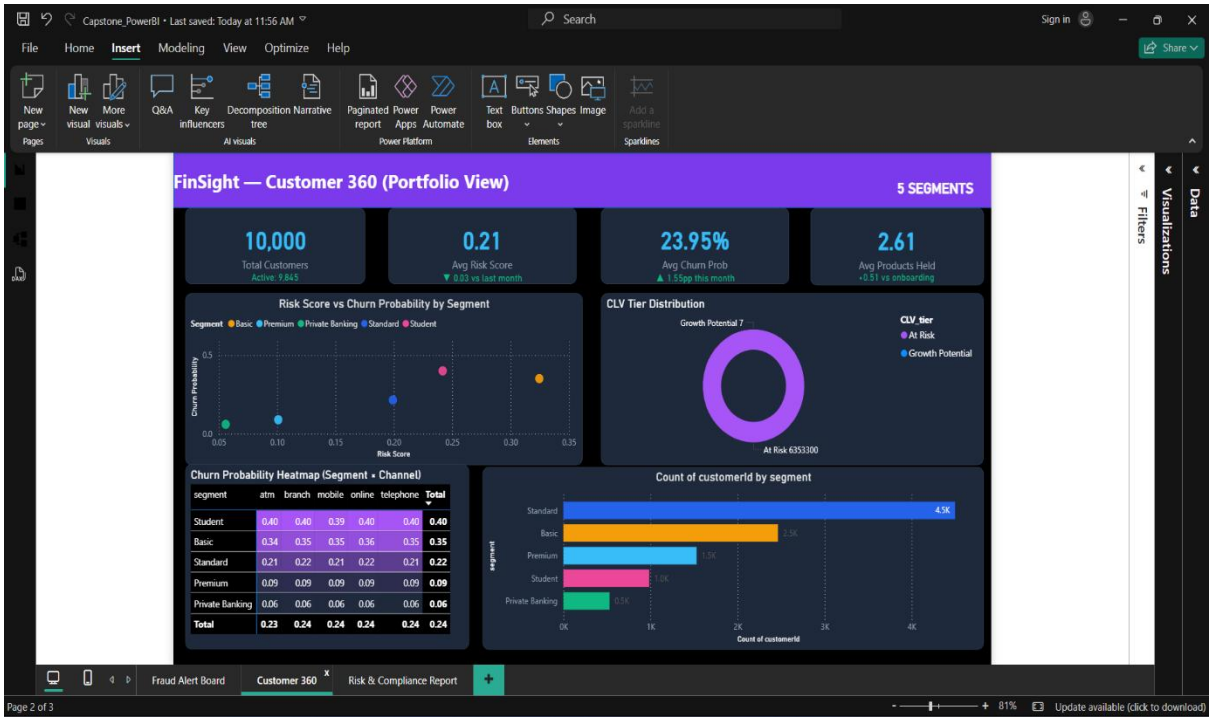
1) Placed a Horizontal Clustered Bar Chart.
2) Set **Y-axis** to segment and **X-axis** to Count of customerId.
3) Applied colour formatting matching the segment colors, enabled data labels.
4) Displays portfolio composition

Step 5: Created Visual 3 — Churn Probability Heatmap (Matrix Table)

1) Selected a Matrix visual
2) Set **Rows** to segment, **Columns** to channel and **Values** to Average of churn_probability.
3) Applied colour formatting to identify which segment and channel combinations carry the highest churn risk

Step 6: Created Visual 4 — CLV Tier Distribution (Donut Chart)

1) Selected a Donut Chart
2) Set **Legend** to CLV_tier and **Values** to Count of customerId.
3) Applied color formatting to the donut slices and styled the container background.
4) Displays customer lifetime value segments (At Risk, Growth Potential)



Dashboard Page 3 — FinSight — Risk & Compliance Report.

Step 1: Header Banner

1) Created the top header banner and added title.
2) Applied background colour formatting.

Step 2: Created DAX Measures & 4 KPI Cards

1) Created DAX measures for Total Transactions, Total Volume, Confirmed Fraud, Dormancy count, and their green subtitle indicators.
2) **Applied Container & Colour Formatting**: Formatted all 4 cards with dark container backgrounds, rounded borders, coloured callout values,

CREATED 4 KPI CARDS:

1) Card 1 (Total Transactions)
2) Card 2 (Total Volume)
3) Card 3 (Confirmed Fraud)
4) Card 4 (Dormant Accounts)

Step 3: Created Transaction Type Filter (Slicer)

1) Selected a Slicer visual

2) Added the field type from compliance_summary and set style to Tile.

3) Applied button styling and container colour formatting with title FILTER BY TRANSACTION TYPE.

4) Allows compliance officers to click any transaction category (CASH_IN, CASH_OUT, DEBIT, PAYMENT, TRANSFER) to instantly filter the entire report.

Step 4: Created Visual 1 — Transaction Volume by Type (Column Chart)

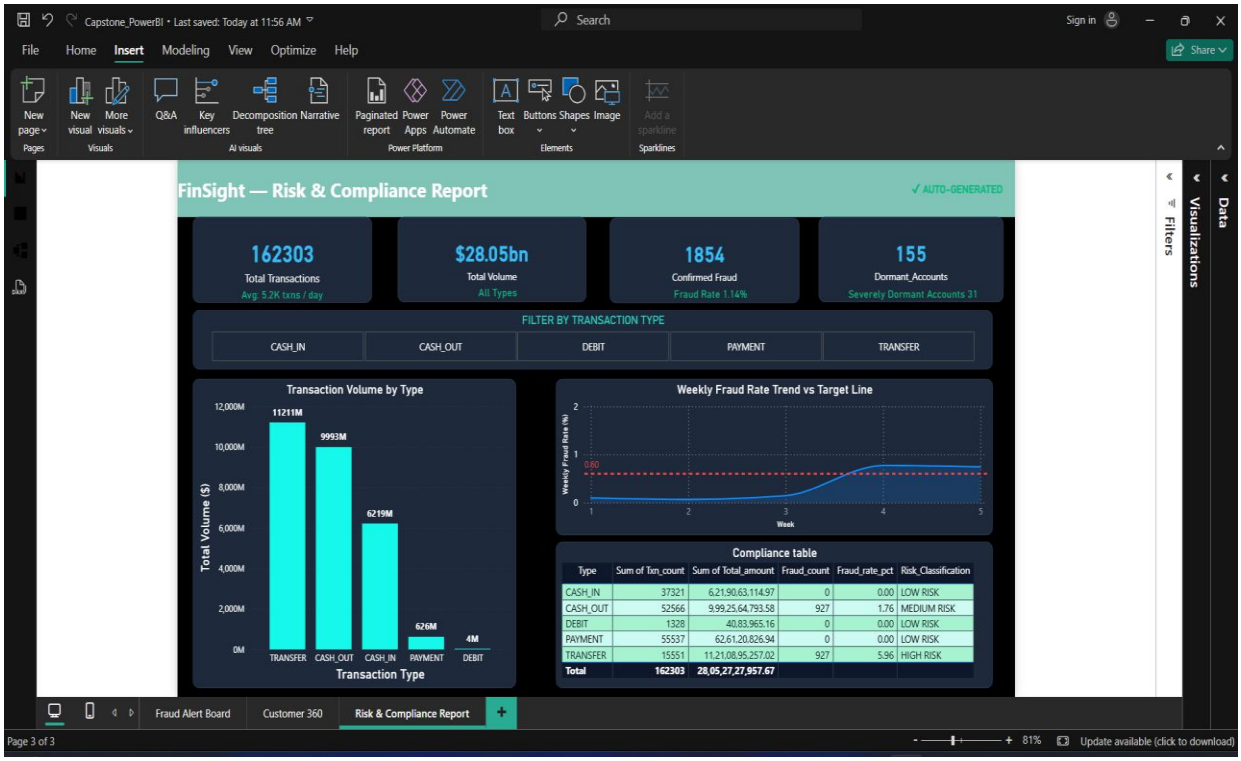
1) Selected a Clustered Column Chart
2) Set X-axis to Transaction Type and Y-axis to Total Volume .
3) Applied colour formatting to the columns**,**

Step 5: Created Visual 2 — Weekly Fraud Rate Trend vs Target Line (Line Chart)

1) Selected a Line Chart
2) Set X-axis to Week (Weeks 1 to 5) and Y-axis to Weekly Fraud Rate (%).
3) Added a constant Target Threshold Line at 0.60% using the Analytics pane.
4) Applied smooth line styling and colour formatting to monitor weekly fraud trajectory and detect when fraud crosses the target ceiling

Step 6: Created Visual 3 — Compliance Table (Audit Data Table)

1) Selected a Table visual
2) Added the 6 required audit columns
3) Added a DAX calculated column for Risk_Classification to classify transaction categories into HIGH RISK, MEDIUM RISK, and LOW RISK.
4) Applied alternating row color formatting, styled headers,



NOTE: The ZIP folder is organized into **Day 1 to Day 5.**

IN ZIP Folder Each day and Each Task Screenshots are uploaded.

NOTE: All Python files (.py), Hive SQL queries, MongoDB queries and Alteryx workflows are provided in their respective task folders.